

CHAPTER 37

The Epidemiology of Infectious Disease



This is a photo of the Centers for Disease Control and Prevention (CDC). The CDC is located next to Emory University in Atlanta, Georgia. It is part of the United States Department of Health and Human Services, Public Health Service (USPHS). In these government facilities, thousands of microbiologists, physicians, statisticians, and social workers study all aspects of infectious human diseases.

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Concepts

1. The science of epidemiology deals with the occurrence and distribution of disease within a given population. Infectious disease epidemiology is concerned with organisms or agents responsible for the spread of infectious diseases in human and other animal populations.
2. Because numbers and time are major epidemiological parameters, statistics is an important working tool in this discipline. Statistics are used to determine morbidity, frequency, and mortality rates.
3. To trace the origin and manner of spread of an infectious disease outbreak, it is necessary to learn what pathogen is responsible.
4. Epidemiologists investigate five links in the infectious disease cycle: (1) characteristics of the pathogen, (2) source and/or reservoir of the pathogen, (3) mode of transmission, (4) susceptibility of the host, and (5) exit mechanisms.
5. Emerging and reemerging diseases and pathogens are a major global concern, as is the potential threat of bioterrorism.
6. Global travel requires global health considerations.
7. The control of nosocomial (hospital acquired) infections has received increasing attention in recent years because of the number of individuals involved, increasing costs, and the length of hospital stays.

Epidemics of infectious disease are often compared with forest fires. Once fire has spread through an area, it does not return until new trees have grown up. Epidemics in humans develop when a large population of susceptible individuals is present. If most individuals are immune, then an epidemic will not occur.

—Andrew Cliff and Peter Haggett

This chapter describes the epidemiological parameters that are studied in the infectious disease cycle. The practical goal of epidemiology is to establish effective recognition, control, prevention, and eradication measures within a given population. Because emerging and reemerging diseases and pathogens as well as the potential threat of bioterrorism are worldwide concerns, these topics are covered here. Global travel requires global health considerations that are continually monitored by epidemiologists. Nosocomial (hospital) acquired infections have increased in recent years, and a brief synopsis of their epidemiology also is presented.

The science of epidemiology originated and evolved in response to the great epidemic diseases such as cholera, typhoid fever, smallpox, and yellow fever (**Box 37.1**). Today its scope encompasses all diseases: infectious diseases and those resulting from anatomical deformities, genetic abnormalities, metabolic dysfunction, malnutrition, neoplasms, psychiatric disorders, and aging. This chapter emphasizes only infectious disease epidemiology.

By definition, **epidemiology** [Greek *epi*, upon, and *demos*, people or population, and *logy*, study] is the science that evaluates the occurrence, determinants, distribution, and control of health and disease in a defined human population. **Health** is the condition in which the organism (and all of its parts) performs its vital functions normally or properly. It is a state of physical and mental well-being and not merely the absence of disease. A **disease** [French *des*, from, and *aise*, ease] is an impairment of the normal

Box 37.1

John Snow: The First Epidemiologist

Much of what we know today about the epidemiology of cholera is based on the classic studies conducted by the British physician John Snow between 1849 and 1854. During this period a series of cholera outbreaks occurred in London, England, and Snow set out to find the source of the disease. Some years earlier when he was still a medical apprentice, Snow had been sent to help during an outbreak of cholera among coal miners. His observations convinced him that the disease was usually spread by unwashed hands and shared food, not by “bad” air or casual direct contact.

Thus when the outbreak of 1849 occurred, Snow believed that cholera was spread among the poor in the same way as among the coal miners. He suspected that water, and not unwashed hands and shared food, was the source of the cholera infection among the wealthier residents. Snow examined official death records and discovered that most of the victims in the Broad Street area had lived close to the Broad Street pump or had been in the habit of drinking from it. He concluded that cholera was spread by drinking water from the Broad Street pump, which was contaminated with raw sewage containing the disease agent. When the pump handle was removed, the number of cholera cases dropped dramatically.

In 1854 another cholera outbreak struck London. Part of the city’s water supply came from two different suppliers: the Southwark and Vauxhall Company and the Lambeth Company. Snow interviewed cholera patients and found that most of them purchased their drinking water from the Southwark and Vauxhall Company. He also discovered that this company obtained its water from the Thames River below locations where Londoners had discharged their sewage. In contrast, the Lambeth Company took its water from the Thames before the river reached the city. The death rate from cholera was over eightfold lower in households supplied with Lambeth Company water. Water contaminated by sewage was transmitting the disease. Finally, Snow concluded that the cause of the disease must be able to multiply in water. Thus he nearly recognized that cholera was caused by a microorganism, though Robert Koch didn’t discover the causative bacterium (*Vibrio cholerae*) until 1883.

To commemorate these achievements, the John Snow Pub now stands at the site of the old Broad Street pump. Those who complete the Epidemiologic Intelligence Program at the Centers for Disease Control and Prevention receive an emblem bearing a replica of a barrel of Watney’s Ale—the brew dispensed at the John Snow Pub.

state of an organism or any of its components that hinders the performance of vital functions. It is a response to environmental factors (e.g., malnutrition, industrial hazards, climate), specific infective agents (e.g., viruses, bacteria, fungi, protozoa, helminths), inherent defects of the body (e.g., various genetic or immunologic anomalies), or combinations of these.

Any individual who practices epidemiology is an **epidemiologist**. Epidemiologists are, in effect, disease detectives. Their major concerns are the discovery of the factors essential to disease occurrence and the development of methods for disease prevention.

37.1 Epidemiological Terminology

When a disease occurs occasionally, and at irregular intervals in a human population, it is a **sporadic disease** (e.g., typhoid fever). When it maintains a steady, low-level frequency at a moderately regular interval, it is an **endemic** [Greek *endemos*, dwelling in the same people] **disease** (e.g., the common cold). **Hyperendemic diseases** gradually increase in occurrence frequency beyond the endemic level but not to the epidemic level (e.g., the common cold during winter months). An **epidemic** [Greek *epidemios*, upon the people] is a sudden increase in the occurrence of a disease above the expected level (**figure 37.1**). Influenza is an example of a disease that often achieves epidemic status. The first case in an epidemic is called the **index case**. An **outbreak**, on the other hand, is the sudden, unexpected occurrence of a disease, usually focally or in a limited segment of a population (e.g., Legionnaires' disease). Although the epidemiology of an outbreak may be no different from that of an epidemic, the community regards the outbreak as less serious. A **pandemic** [Greek *pan*, all] is an increase in disease occurrence within a large population over a very wide region (usually the world). Usually, pandemic diseases spread among continents. The influenza outbreak of the 1960s and AIDS in the 1980s are good examples.

The discipline that deals with the factors that influence the frequency of a disease in an animal population is known as **epizootiology**. Moderate prevalence of a disease in animals is termed **enzootic**, a sudden outbreak is **epizootic**, and wide dissemination is **panzootic**. Animal diseases that can be transmitted to humans are termed **zoonoses** [Greek *zoon*, animal, and *nosos*, disease].

37.2 Measuring Frequency: The Epidemiologist's Tools

An important tool used in the study of modern epidemiology is statistics. **Statistics** is the branch of mathematics dealing with the collection, organization, and interpretation of numerical data. As a science particularly concerned with rates and the comparison of rates, epidemiology was the first medical field in which statistical methods were extensively used.

Measures of frequency usually are expressed as fractions. The numerator is the number of individuals experiencing the event—infection or other problem—and the denominator is the number of individuals in whom the event could have occurred, that is, the pop-

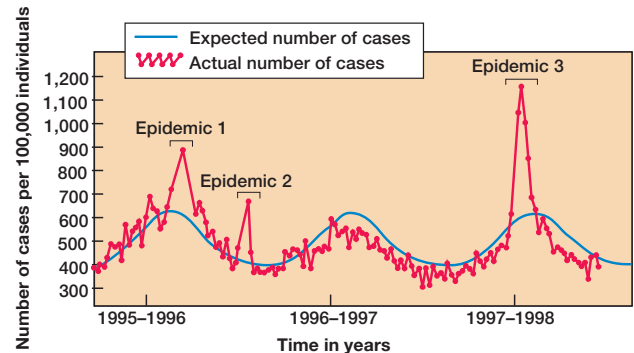


Figure 37.1 A Graph Illustrating Three Epidemics. The solid blue line indicates the expected number of endemic cases. The connected red dots indicate the actual number of cases. Epidemics (marked by brackets) are sharp increases in the number of cases of a disease above that which is normally expected (solid line).

ulation at risk. The fraction is a proportion or ratio but is commonly called a rate because a time period is always specified. (A rate also can be expressed as a percentage.) In population statistics, rates usually are stated per 1,000 individuals, although other powers of 10 may be used for particular diseases (e.g., per 100 for very common diseases and per 10,000 or 100,000 for uncommon diseases).

A **morbidity rate** measures the number of individuals that become ill due to a specific disease within a susceptible population during a specific time interval. It is an incidence rate and reflects the number of new cases in a period. The rate is commonly determined when the number of new cases of illness in the general population is known from clinical reports. It is calculated as follows:

$$\text{Morbidity rate} = \frac{\text{number of new cases of a disease during a specific period}}{\text{number of individuals in the population}}$$

For example, if there were 700 new cases of influenza per 100,000 individuals, then the morbidity rate would be expressed as 700 per 100,000 or 0.7%.

The **prevalence rate** refers to the total number of individuals infected in a population at any one time no matter when the disease began. The prevalence rate depends on both the incidence rate and the duration of the illness.

The **mortality rate** is the relationship of the number of deaths from a given disease to the total number of cases of the disease. The mortality rate is a simple statement of the proportion of all deaths that are assigned to a single cause. It is calculated as follows:

$$\text{Mortality rate} = \frac{\text{number of deaths due to a given disease}}{\text{size of the total population with the same disease}}$$

For example, if there were 15,000 deaths due to AIDS in a year, and the total number of people infected was 30,000, the mortality rate would be 15,000 per 30,000 or 1 per 2 or 50%.

The determination of morbidity, prevalence, and mortality rates aids public health personnel in directing health-care efforts to control the spread of infectious diseases. For example, a sudden

increase in the morbidity rate of a particular disease may indicate a need for the implementation of preventive measures designed to reduce mortality.

1. What is epidemiology?
2. What terms are used to describe the occurrence of a disease in a human population? In an animal population?
3. Define morbidity rate, prevalence rate, and mortality rate.
4. How would you define a disease? Health?

37.3 Infectious Disease Epidemiology

An infectious disease is a disease resulting from an infection by microbial agents such as viruses, bacteria, fungi, protozoa, and helminths that can be transmitted from one host to another. The manifestations of an infectious disease can range from mild to severe to deadly depending on the agent and host. An epidemiologist studying an infectious disease is concerned with the causative agent, the source and/or reservoir of the disease agent (p. 854), how it was transmitted, what host and environmental factors could have aided development of the disease within a defined population, and how best to control or eliminate the disease. These factors describe the natural history or cycle of an infectious disease.

37.4 Recognition of an Infectious Disease in a Population

Epidemiologists can recognize an infectious disease in a population by using various surveillance methods. Surveillance is a dynamic activity that includes gathering information on the development and occurrence of a disease, collating and analyzing the data, summarizing the findings, and using the information to select control methods. Some combination of the following surveillance methods is used most often:

1. Generation of morbidity data from case reports
2. Collection of mortality data from death certificates
3. Investigation of actual cases
4. Collection of data from reported epidemics
5. Field investigation of epidemics
6. Review of laboratory results: surveys of a population for antibodies against the agent and specific microbial serotypes, skin tests, cultures, stool analyses, etc.
7. Population surveys using valid statistical sampling to determine who has the disease
8. Use of animal and vector disease data
9. Collection of information on the use of specific biologics—antibiotics, antitoxins, vaccines, and other prophylactic measures
10. Use of demographic data on population characteristics such as human movements during a specific time of the year
11. Use of remote sensing and geographic information systems

As noted, surveillance may not always require the direct examination of cases. However, to accurately interpret surveillance data and study the course of a disease in individuals, epidemiologists and other medical professionals must be aware of the pattern of infectious diseases. Often infectious diseases have characteristic signs and symptoms. **Signs** are objective changes in the body, such as a fever or rash, that can be directly observed. **Symptoms** are subjective changes, such as pain and loss of appetite, that are personally experienced by the patient. The term symptom is often used in a broader scope to include the clinical signs. A **disease syndrome** is a set of signs and symptoms that are characteristic of the disease. Frequently additional laboratory tests are required for an accurate diagnosis because symptoms and readily observable signs are not sufficient for diagnosis.

The course of an infectious disease usually has a characteristic pattern and can be divided into several phases. Knowledge of the pattern is essential in accurately diagnosing the disease.

1. The **incubation period** is the period between pathogen entry and the expression of signs and symptoms. The pathogen is spreading but has not reached a sufficient level to cause clinical manifestations. This period's length varies with disease.
2. The **prodromal stage** is the period in which there is an onset of signs and symptoms, but they are not yet specific enough to make a diagnosis. The patient often is contagious.
3. The illness period is the phase in which the disease is most severe and has characteristic signs and symptoms. The immune response has been triggered; B and T cells are becoming active.
4. In the period of decline, the signs and symptoms begin to disappear. The recovery stage often is referred to as convalescence.

Remote Sensing and Geographic Information Systems: Charting Infectious Diseases

Remote sensing and geographic information systems are map-based tools that can be used to study the distribution, dynamics, and environmental correlates of microbial diseases. **Remote sensing (RS)** is the gathering of digital images of the Earth's surface from satellites and transforming the data into maps. A **geographic information system (GIS)** is a data management system that organizes and displays digital map data from RS and facilitates the analysis of relationships between mapped features. Statistical relationships often exist between mapped features and diseases in natural host or human populations. Examples include the location of the habitats of the malaria parasite and mosquito vectors in Mexico and Asia, Rift Valley fever in Kenya, Lyme disease in the United States, and African trypanosomiasis and schistosomiasis in both humans and livestock in the southeastern United States. RS and GIS may also permit the assessment of human risk from pathogens such as Sin Nombre virus (the virus that causes hantavirus pulmonary syndrome in North America). RS and GIS are most useful if disease dynamics and distributions are clearly related to mapped environmental variables. For example, if a mi-

Box 37.2

“Typhoid Mary”

In the early 1900s there were thousands of typhoid fever cases, and many died of the disease. Most of these cases arose when people drank water contaminated with sewage or ate food handled by or prepared by individuals who were shedding the typhoid fever bacterium (*Salmonella typhi*). The most famous carrier of the typhoid bacterium was Mary Mallon.

Between 1896 and 1906 Mary Mallon worked as a cook in seven homes in New York City. Twenty-eight cases of typhoid fever occurred in these homes while she worked in them. As a result the New York City Health Department had Mary arrested and admitted to an isolation hospital on North Brother Island in New York’s East River. Examination of

Mary’s stools showed that she was shedding large numbers of typhoid bacteria though she exhibited no external symptoms of the disease. An article published in 1908 in the *Journal of the American Medical Association* referred to her as “Typhoid Mary,” an epithet by which she is still known today. After being released when she pledged not to cook for others or serve food to them, Mary changed her name and began to work as a cook again. For five years she managed to avoid capture while continuing to spread typhoid fever. Eventually the authorities tracked her down. She was held in custody for 23 years until she died in 1938. As a lifetime carrier, Mary Mallon was positively linked with 10 outbreaks of typhoid fever, 53 cases, and 3 deaths.

crobial disease is associated with certain vegetation types or physical characteristics (elevation, precipitation), RS and GIS can identify regions where risk is relatively high.

Correlation with a Single Causative Agent

After an infectious disease has been recognized in a population, epidemiologists correlate the disease outbreak with a specific organism—its exact cause must be discovered (Box 37.2). At this point the clinical or diagnostic microbiology laboratory enters the investigation. Its purpose is to isolate and identify the organism responsible for the disease.

37.5 Recognition of an Epidemic

As previously noted, an infectious disease epidemic is usually a short-term increase in the occurrence of the disease in a particular population (figure 37.1). Two major types of epidemic are recognized: common source and propagated.

A **common-source epidemic** is characterized as having reached a peak level within a short period of time (1 to 2 weeks) followed by a moderately rapid decline in the number of infected patients (figure 37.2a). This type of epidemic usually results from a single common contaminated source such as food (food poisoning) or water (Legionnaires’ disease).

A **propagated epidemic** is characterized by a relatively slow and prolonged rise and then a gradual decline in the number of individuals infected (figure 37.2b). This type of epidemic usually results from the introduction of a single infected individual into a susceptible population. The initial infection is then propagated to others in a gradual fashion until many individuals within the population are infected. An example is the increase in mumps or chickenpox cases that coincides with new populations of sensitive children who arrive in classrooms each fall. Only one infected child is necessary to initiate the epidemic.

To understand how epidemics are propagated, consider figure 37.3. At time 0, all individuals in this population are suscep-

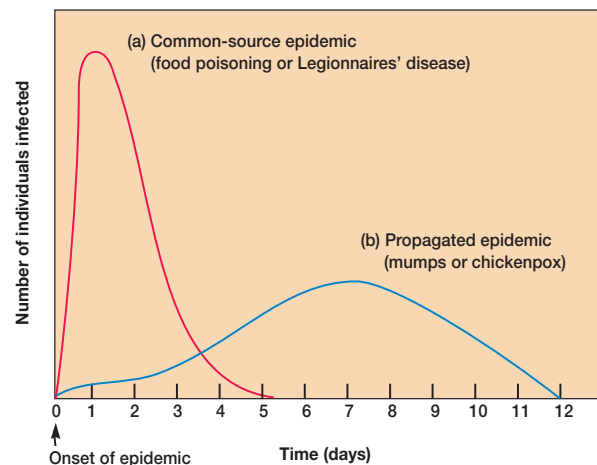


Figure 37.2 Epidemic Curves. (a) In a common-source epidemic, there is a rapid increase up to a peak in the number of individuals infected and then a rapid but more gradual decline. Cases usually are reported for a period that equals approximately one incubation period of the disease. (b) In a propagated epidemic the curve has a gradual rise and then a gradual decline. Cases usually are reported over a time interval equivalent to several incubation periods of the disease.

tible to a hypothetical pathogen. The introduction of an infected individual initiates the epidemic outbreak (lower curve), which spreads and reaches a peak (day 15). As individuals recover from the disease, they become immune and no longer transmit the pathogen (upper curve). The number of susceptible individuals therefore decreases. The decline in the number of susceptibles to the threshold density (the minimum number of individuals necessary to continue propagating the disease) coincides with the peak of the epidemic wave, and the incidence of new cases declines because the pathogen cannot propagate itself.

Herd immunity is the resistance of a population to infection and pathogen spread because of the immunity of a large percentage

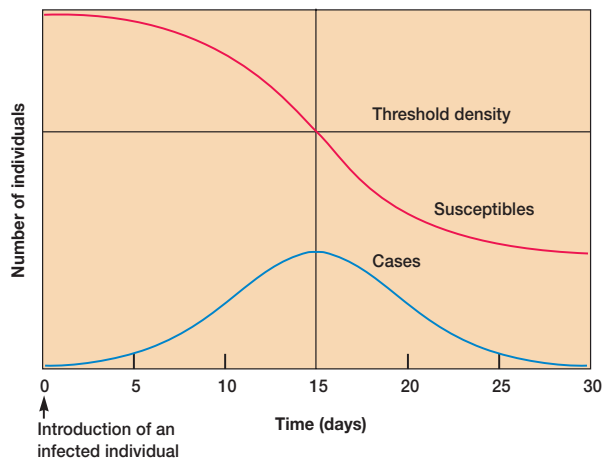


Figure 37.3 Diagrammatic Representation of the Spread of an Imaginary Propagated Epidemic. The lower curve represents the number of cases and the upper curve the number of susceptible individuals. Notice the coincidence of the peak of the epidemic wave with the threshold density of susceptibles.

of the population (figure 37.4). The larger the proportion of those immune, the smaller the probability of effective contact between infective and susceptible individuals—that is, many contacts will be with immunes, and thus the population will exhibit a group resistance. A susceptible member of such an immune population enjoys an immunity that is not of his or her own making (not self-made) but instead arises because of membership in the group.

At times public health officials immunize large portions of the susceptible population in an attempt to maintain a high level of herd immunity. Any increase in the number of susceptibles may result in an endemic disease becoming epidemic. The proportion of immune to susceptible individuals must be constantly monitored because new susceptible individuals continually enter a population through migration and birth. In addition, pathogens can change so much through processes such as antigenic shift (see next paragraph) that immune individuals become susceptible again.

Many pathogens do not ordinarily change in nature. They cause endemic diseases because infected humans continually transfer them to others (e.g., sexually transmitted diseases) or because they continually reenter the human population from animal reservoirs (e.g., rabies). Other pathogens continue to evolve and may produce epidemics (e.g., AIDS, influenza virus [A strain], and *Legionella* bacteria). One way in which a pathogen changes is by **antigenic shift**, a major genetically determined change in the antigenic character of a pathogen. An antigenic shift can be so extensive that the pathogen is no longer recognized by the host's immune system. For example, influenza viruses frequently change by recombination from one antigenic type to another. Antigenic shift also occurs through the hybridization of different influenza virus serovars; two serovars of a virus intermingle to form a new antigenic type. Hybridization may occur between an animal strain and a human strain of the virus. Even though resist-

ance in the human population becomes so high that the virus can no longer spread (herd immunity), it can be transmitted to animals, where the hybridization takes place. Smaller antigenic changes also can take place by a mutation in pathogen strains and help the pathogen avoid host immune responses. These smaller changes are called **antigenic drift**.

Whenever antigenic shift or drift occurs, the population of susceptibles increases because the immune system has not been exposed to the new mutant strain. If the percentage of susceptibles is above the threshold density (figure 37.3), the level of protection provided by herd immunity will decrease and the morbidity rate will increase. For example, the morbidity rates of diphtheria and measles among school children may reach epidemic levels if the number of susceptibles rises above 30% for the whole population. As a result the goal of public health agencies is to make sure that at least 70% of the population is immunized against these diseases to provide the herd immunity necessary for protection of those who are not immunized.

1. How can epidemiologists recognize an infectious disease in a population? Define sign, symptom, and disease syndrome. What are the four phases seen during the course of an infection?
2. How can remote sensing and geographic information systems chart infectious diseases?
3. Differentiate between common-source and propagated epidemics.
4. Explain herd immunity.
5. What is the significance of antigenic shift and drift in epidemiology?

37.6 The Infectious Disease Cycle: Story of a Disease

To continue to exist, a pathogen must reproduce and be disseminated among its hosts. Thus an important aspect of infectious disease epidemiology is a consideration of how reproduction and dissemination occur. The **infectious disease cycle** or **chain of infection** represents these events in the form of an intriguing epidemiological mystery story (figure 37.5).

What Pathogen Caused the Disease?

The first link in the infectious disease cycle is the pathogen. After an infectious disease has been recognized in a population, epidemiologists must correlate the disease outbreak with a specific pathogen. The disease's exact cause must be discovered. This is where Koch's postulates (see section 1.3), and modifications of them, are used to determine the etiology or cause of an infectious disease. At this point the clinical or diagnostic microbiology laboratory enters the investigation (see section 36.1). Its purpose is to isolate and identify the pathogen that caused the disease and to determine the pathogen's susceptibility to antimicrobial agents or methods that may assist in its eradication.

Many pathogens can cause infectious diseases in humans and will be discussed in detail in chapters 38 to 40. Often these pathogens are transmissible from one individual to another. A

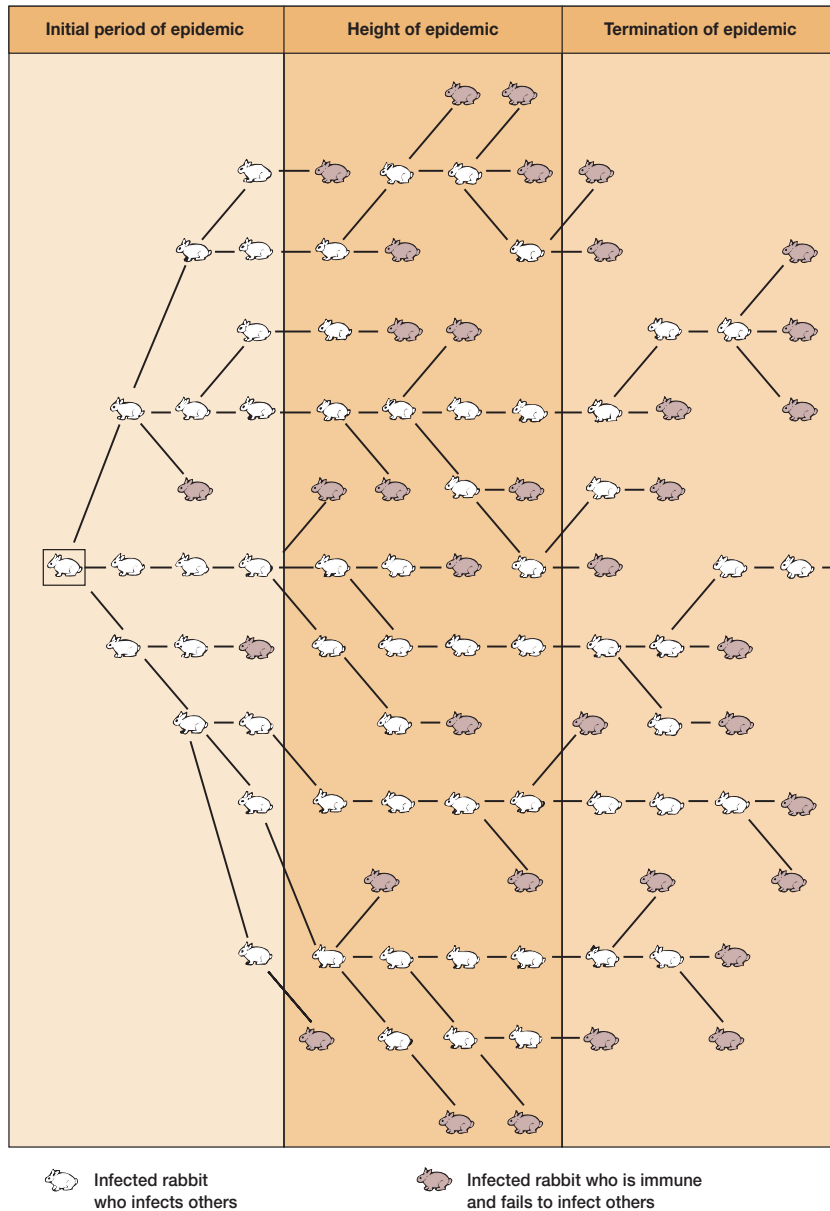


Figure 37.4 Herd Immunity. The kinetics of the spread of an infectious disease and the effect of increasing the number of immune individuals in the population in limiting the disease. On day 1, a single infected individual enters the population. The incubation period is 1 day, and recovery occurs in 2 days. The number of susceptibles is the total population on day 1. The number of infected and recovered are illustrated in the two graphs.

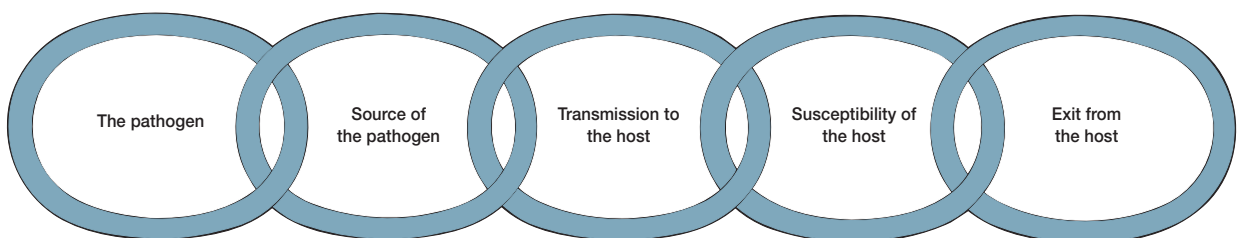
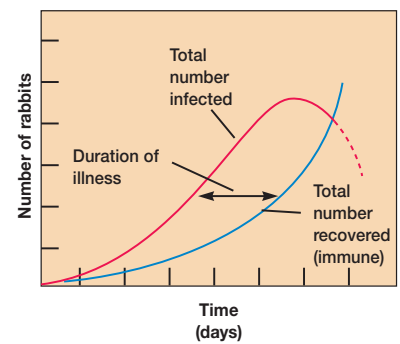
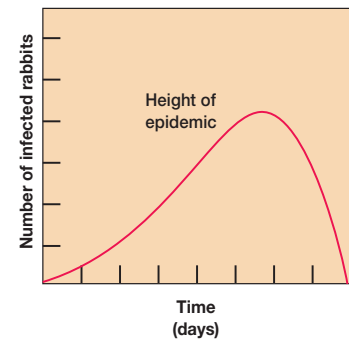


Figure 37.5 Infectious Disease Cycle or Chain of Infection. See text for further details.

Table 37.1 Infectious Organisms in Nonhuman Reservoirs That May Be Transmitted to Humans

Disease	Etiologic Agent	Usual or Suspected Nonhuman Host	Usual Method of Human Infection
Acquired human LCMV infection	Lymphocytic choriomeningitis virus	Viremic mice or hamsters	Contact with infected rodents
Anthrax	<i>Bacillus anthracis</i>	Cattle, horses, sheep, swine, goats, dogs, cats, wild animals, birds	Inhalation or ingestion of spores; direct contact
Babesiosis	<i>Babesia bovis</i> , <i>B. divergens</i> , <i>B. microti</i> , <i>B. equi</i>	<i>Ixodes</i> ticks of various species	Bite of infected tick
Brucellosis (undulant fever)	<i>Brucella melitensis</i> , <i>B. abortus</i> , <i>B. suis</i>	Cattle, goats, swine, sheep, horses, mules, dogs, cats, fowl, deer, rabbits	Milk; direct or indirect contact
Campylobacteriosis	<i>Campylobacter fetus</i> , <i>C. jejuni</i>	Cattle, sheep, poultry, swine, pets, other animals	Contaminated water and food
Cat-scratch disease	<i>Bartonella henselae</i>	Cats, dogs	Cat or dog scratch
Colorado tick fever	<i>Coltivirus</i>	Squirrels, chipmunks, mice, deer	Tick bite
Cowpox	Cowpox virus	Cattle, horses	Skin abrasions
Cryptosporidiosis	<i>Cryptosporidium</i> spp.	Calves	Contact with infected calves
Encephalitis (California)	Arbovirus	Rats, squirrels, horses, deer, hares, cows	Mosquito
Encephalitis (St. Louis)	Arbovirus	Birds	Mosquito
Encephalomyelitis (Eastern equine)	Arbovirus	Birds, ducks, fowl, horses	Mosquito
Encephalomyelitis (Venezuelan equine)	Arbovirus	Rodents, horses	Mosquito
Encephalomyelitis (Western equine)	Arbovirus	Birds, snakes, squirrels, horses	Mosquito
Giardiasis	<i>Giardia lamblia</i>	Rodents, deer, cattle, dogs, cats	Contaminated water
Glanders	<i>Pseudomonas mallei</i>	Horses	Skin contact; inhalation
Hantavirus pulmonary syndrome	Pulmonary syndrome hantavirus	Deer mice	Contact with the saliva, urine, or feces of deer mice; aerosolized viruses
Herpes B viral encephalitis	<i>Herpesvirus simiae</i>	Monkeys	Monkey bite; contact with material from monkeys
Leptospirosis	<i>Leptospira interrogans</i>	Dogs, rodents, wild animals	Direct contact with urine, infected tissue, and contaminated water
Listeriosis	<i>Listeria monocytogenes</i>	Sheep, cattle, goats, guinea pigs, chickens, horses, rodents, birds, crustaceans	Food-borne
Lyme disease	<i>Borrelia burgdorferi</i>	Ticks (<i>Ixodes scapularis</i>) or related ticks	Bite of infected tick
Lymphocytic choriomeningitis	Arbovirus	Mice, rats, dogs, monkeys, guinea pigs	Inhalation of contaminated dust; ingestion of contaminated food

communicable disease is an illness caused by a pathogen or its products that has been transmitted from an infected person or a reservoir, either directly or indirectly. Pathogens have the potential to produce disease (pathogenicity); this potential is a function of such factors as the number of pathogens, their virulence, and the nature and magnitude of host defenses.

What Was the Source and/or Reservoir of the Pathogen?

The source and/or reservoir of a pathogen is the second link in the infectious disease cycle. Identifying the source and/or reservoir is an important aspect of epidemiology. If the source or reservoir of the infection can be eliminated or controlled, the infectious disease cycle itself will be interrupted and transmission of the pathogen will be prevented (Boxes 37.1 and 37.2).

A **source** is the location from which the pathogen is immediately transmitted to the host, either directly through the environment or indirectly through an intermediate agent. The source can be either animate (e.g., humans or animals) or inanimate (e.g., water, soil, or food). The **period of infectivity** is the time during which the source is infectious or is disseminating the pathogen.

The **reservoir** is the site or natural environmental location in which the pathogen is normally found living and from which infection of the host can occur. Thus a reservoir sometimes functions as a source. Reservoirs also can be animate or inanimate.

Much of the time, human hosts are the most important animate sources of the pathogen and are called carriers. A **carrier** is an infected individual who is a potential source of infection for

Table 37.1 Continued

Disease	Etiologic Agent	Usual or Suspected Nonhuman Host	Usual Method of Human Infection
Mediterranean fever (boutonneuse fever, African tick typhus)	<i>Rickettsia conorii</i>	Dogs	Tick bite
Melioidosis	<i>Pseudomonas pseudomallei</i>	Rats, mice, rabbits, dogs, cats	Arthropod vectors, water, food
Orf (contagious ecthyma)	Virus	Sheep, goats	Through skin abrasions
Pasteurellosis	<i>Pasteurella multocida</i>	Fowl, cattle, sheep, swine, goats, mice, rats, rabbits	Animal bite
Plague (bubonic)	<i>Yersinia pestis</i>	Domestic rats, many wild rodents	Flea bite
Psittacosis	<i>Chlamydia psittaci</i>	Birds	Direct contact, respiratory aerosols
Q fever	<i>Coxiella burnetii</i>	Cattle, sheep, goats	Inhalation of infected soil and dust
Rabies	Rabies virus (Rhabdovirus group)	Dogs, bats, opossums, skunks, foxes, cats, cattle	Bite of rabid animal
Rat bite fever	<i>Spirillum minus</i> <i>Streptobacillus moniliformis</i>	Rats, mice, cats	Rat bite
Relapsing fever (borreliosis)	<i>Borrelia</i> spp.	Rats, mice, squirrels, weasels, turkeys, contaminated food	Rat bite
Rickettsialpox	<i>Rickettsia akari</i>	Rodents, porcupines, opossums, armadillos, ticks, lice	Tick or louse bite
Rocky Mountain spotted fever	<i>Rickettsia rickettsii</i>	Mice	Mite bite
Salmonellosis	<i>Salmonella</i> spp. (except <i>S. typhosa</i>)	Rabbits, squirrels, rats, mice, groundhogs	Tick bite
Scrub typhus	<i>Rickettsia tsutsugamushi</i>	Fowl, swine, sheep, cattle, horses, dogs, cats, rodents, reptiles, birds, turtles	Direct contact; food
Tuberculosis	<i>Mycobacterium bovis</i>	Wild rodents, rats	Mite bite
Tularemia	<i>Francisella tularensis</i>	Cattle, horses, cats, dogs	Milk; direct contact
Typhus fever (endemic)	<i>Rickettsia mooseri</i>	Wild rabbits, most other wild and domestic animals	Direct contact with infected carcass, usually rabbit; tick bite, biting flies
Vesicular stomatitis	Virus (Rhabdovirus group)	Rats	Flea bite
Weil's disease (leptospirosis)	<i>Leptospira interrogans</i>	Cattle, swine, horses	Direct contact
Yellow fever (jungle)	Yellow fever virus	Rats, mice, skunks, opossums, wildcats, foxes, raccoons, shrews, bandicoots, dogs, cattle, swine	Through skin, drinking water, eating food
		Monkeys, marmosets, lemurs, mosquitoes	Mosquito

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others. Carriers play an important role in the epidemiology of disease. Four types of carriers are recognized:

1. An **active carrier** is an individual who has an overt clinical case of the disease.
2. A **convalescent carrier** is an individual who has recovered from the infectious disease but continues to harbor large numbers of the pathogen.
3. A **healthy carrier** is an individual who harbors the pathogen but is not ill.
4. An **incubatory carrier** is an individual who is incubating the pathogen in large numbers but is not yet ill.

Convalescent, healthy, and incubatory carriers may harbor the pathogen for only a brief period (hours, days, or weeks) and then are called **casual, acute, or transient carriers**. If they harbor the pathogen for long periods (months, years, or life), they are called **chronic carriers**.

As noted earlier, infectious diseases called zoonoses occur in animals and are occasionally transmitted to humans; thus these animals also can serve as reservoirs. Humans contract the pathogen by several mechanisms: coming into direct contact with diseased animal flesh (tularemia); drinking contaminated cow's milk (tuberculosis and brucellosis); inhaling dust particles contaminated by animal excreta or products (Q fever, anthrax); or eating insufficiently cooked infected flesh (anthrax, trichinosis). In addition, being bitten by arthropod **vectors** (organisms that spread disease from one host to another) such as mosquitoes, ticks, fleas, mites, or biting flies (equine encephalomyelitis and malaria, Lyme disease, Rocky Mountain spotted fever, plague, scrub typhus, and tularemia); or being bitten by a diseased animal (rabies) can lead to infection.

Table 37.1 lists some more common zoonoses found in the Western Hemisphere. This table is noninclusive in scope; it merely abbreviates the enormous spectrum of zoonotic diseases that are relevant to human epidemiology. Domestic animals are



Figure 37.6 A Sneeze. High-speed photograph of an aerosol generated by an unstifled sneeze. The particles seen are comprised of saliva and mucus laden with microorganisms. These airborne particles may be infectious when inhaled by a susceptible host. Even a surgical mask will not prevent the spread of all particles.

the most common source of zoonoses because they live in greater proximity to humans than do wild animals. Diseases of wild animals that are transmitted to humans tend to occur sporadically because close contact is infrequent. Other major reservoirs of pathogens are water, soil, and food. These reservoirs are discussed in detail in chapters 29, 30, and 41.

How Was the Pathogen Transmitted?

To maintain an active infectious disease in a human population, the pathogen must be transmitted from one host or source to another. Transmission is the third link in the infectious disease cycle and occurs by four main routes: airborne, contact, vehicle, and vector-borne.

Airborne Transmission

Because air is not a suitable medium for the growth of a pathogen, any pathogen that is airborne must have originated from a source such as humans, other animals, plants, soil, food, or water. In **airborne transmission** the pathogen is truly suspended in the air and travels over a meter or more from the source to the host. The pathogen can be contained within droplet nuclei or dust. **Droplet nuclei** are small particles, 1 to 4 μm in diameter, that result from the evaporation of larger particles (10 μm or more in diameter) called droplets. Droplet nuclei can remain airborne for hours or days and travel long distances.

When animals or humans are the source of the airborne pathogen, it usually is propelled from the respiratory tract into the air by an individual's coughing, sneezing, or vocalization. For example, enormous numbers of moisture droplets are aerosoled

during a typical sneeze (**figure 37.6**). Each droplet is about 10 μm in diameter and initially moves about 100 m/second or more than 200 mi/hour!

Dust also is an important route of airborne transmission. At times a pathogen adheres to dust particles and contributes to the number of airborne pathogens when the dust is resuspended by some disturbance. A pathogen that can survive for relatively long periods in or on dust creates an epidemiological problem, particularly in hospitals, where dust can be the source of hospital-acquired infections. **Table 37.2** summarizes some human airborne pathogens and the diseases they cause.

Contact Transmission

Contact transmission implies the coming together or touching of the source or reservoir of the pathogen and the host (**Box 37.3**). Contact can be direct, indirect, or by droplet spread. Direct contact implies an actual physical interaction with the infectious source. This route is frequently called person-to-person contact. Person-to-person transmission occurs primarily by touching, kissing, or sexual contact (sexually transmitted diseases); by contact with oral secretions or body lesions (herpes and boils); by nursing mothers (staphylococcal infections); and through the placenta (AIDS, syphilis). Some infectious pathogens also can be transmitted by direct contact with animals or animal products (*Salmonella* and *Campylobacter*).

Indirect contact refers to the transmission of the pathogen from the source to the host through an intermediary—most often an inanimate object. The intermediary is usually contaminated by an animate source. Common examples of intermediary inanimate objects include thermometers, eating utensils, drinking cups, and

Table 37.2 Some Airborne Pathogens and the Diseases They Cause in Humans

Microorganism	Disease	Microorganism	Disease
Viruses		Bacteria	
Varicella	Chickenpox	<i>Actinomyces</i> spp.	Lung infections
Influenza	Flu	<i>Bordetella pertussis</i>	Whooping cough
Rubeola	Measles	<i>Chlamydia psittaci</i>	Psittacosis
Rubella	German measles	<i>Corynebacterium diphtheriae</i>	Diphtheria
Mumps	Mumps	<i>Mycoplasma pneumoniae</i>	Pneumonia
Poliomyelitis	Polio	<i>Mycobacterium tuberculosis</i>	Tuberculosis
Acute respiratory viruses	Viral pneumonia	<i>Neisseria meningitidis</i>	Meningitis
Pulmonary syndrome hantavirus	Hantavirus pulmonary syndrome	<i>Streptococcus</i> spp.	Pneumonia, sore throat
		Fungi	
		<i>Blastomyces</i> spp.	Lung infections
		<i>Coccidioides</i> spp.	Coccidioidomycosis
		<i>Histoplasma capsulatum</i>	Histoplasmosis

Box 37.3**The First Indications of Person-to-Person Spread of an Infectious Disease**

In 1773 Charles White, an English surgeon and obstetrician, published his “Treatise on the Management of Pregnant and Lying-In Women.” In it, he appealed for surgical cleanliness to combat childbed or puerperal fever. (**Puerperal fever** is an acute febrile condition that can follow childbirth and is caused by streptococcal infection of the uterus and/or adjacent regions.) In 1795 Alexander Gordon, a Scottish obstetrician, published his “Treatise on the Epidemic Puerperal Fever of Aberdeen,” which demonstrated for the first time the contagiousness of the disease. In 1843 Oliver Wendell Holmes, a noted physician and anatomist in the United States, published a paper entitled “On the Contagiousness of Puerperal Fever” and also appealed for surgical cleanliness to combat this disease.

However, the first person to realize that a pathogen could be transmitted from one person to another was the Hungarian physician Ignaz Phillip Semmelweis. Between 1847 and 1849 Semmelweis observed that

women who had their babies at the hospital with the help of medical students and physicians were four times as likely to contract puerperal fever as those who gave birth with the help of midwives. He concluded that the physicians and students were infecting women with material remaining on their hands after autopsies and other activities. Semmelweis thus began washing his hands with a calcium chloride solution before examining patients or delivering babies. This simple procedure led to a dramatic decrease in the number of cases of puerperal fever and saved the lives of many women. As a result Semmelweis is credited with being the pioneer of antiseptic in obstetrics. Unfortunately, in his own time, most of the medical establishment refused to acknowledge his contribution and adopt his procedures. After years of rejection Semmelweis had a nervous breakdown in 1865. He died a short time later of a wound infection. It is very probable that it was a streptococcal infection, arising from the same pathogen he had struggled against his whole professional life.

bedding. *Pseudomonas* bacteria are easily transmitted by this route. This mode of transmission is often also considered a form of vehicle transmission (see next section).

In droplet spread the pathogen is carried on particles larger than 5 μm . The route is through the air but only for a very short distance—usually less than a meter. Because these particles are large, they quickly settle out of the air. As a result droplet transmission of a pathogen depends on the proximity of the source and the host. Measles is an example of a droplet-spread disease.

Vehicle Transmission

Inanimate materials or objects involved in pathogen transmission are called **vehicles**. In **common vehicle transmission** a single inanimate vehicle or source serves to spread the pathogen

to multiple hosts but does not support its reproduction. Examples include surgical instruments, bedding, and eating utensils. In epidemiology these common vehicles are called **fomites** [s., fomes or fomite]. A single source containing pathogens (blood, drugs, IV fluids) can contaminate a common vehicle that causes multiple infections. Food and water are important common vehicles for many human diseases (see tables 39.5 and 41.6).

Vector-Borne Transmission

As noted earlier, living transmitters of a pathogen are called vectors. Most vectors are arthropods (insects, ticks, mites, fleas) or vertebrates (dogs, cats, skunks, bats). **Vector-borne transmission** can be either external or internal. In external (mechanical)

transmission the pathogen is carried on the body surface of a vector. Carriage is passive, with no growth of the pathogen during transmission. An example would be flies carrying *Shigella* organisms on their feet from a fecal source to a plate of food that a person is eating.

In internal transmission the pathogen is carried within the vector. Here it can go into either a harborage or biologic transmission phase. In **harborage transmission** the pathogen does not undergo morphological or physiological changes within the vector. An example would be the transmission of *Yersinia pestis* (the etiologic agent of plague) by the rat flea from rat to human. **Biologic transmission** implies that the pathogen does go through a morphological or physiological change within the vector. An example would be the developmental sequence of the malarial parasite inside its mosquito vector. [Malaria](#) (pp. 954–55)

Why Was the Host Susceptible to the Pathogen?

The fourth link in the infectious disease cycle is the host. The susceptibility of the host to a pathogen depends on both the pathogenicity of the organism and the nonspecific and specific defense mechanisms of the host. These susceptibility factors are the basis for chapters 31 and 32 and are outside the realm of epidemiology.

How Did the Pathogen Leave the Host?

The fifth and last link in the infectious disease cycle is release or exit of the pathogen from the host. It is equally important that the pathogen escapes from its host as it is that the pathogen originally contacts and enters the host. Unless a successful escape occurs, the disease cycle will be interrupted and the pathogenic species will not be perpetuated. Escape can be active or passive, although often a combination of the two occurs. Active escape takes place when a pathogen actively moves to a portal of exit and leaves the host. Examples include the many parasitic helminths that migrate through the body of their host, eventually reaching the surface and exiting. Passive escape occurs when a pathogen or its progeny leaves the host in feces, urine, droplets, saliva, or desquamated cells. Microorganisms usually employ passive escape mechanisms.

1. What are some epidemiologically important characteristics of a pathogen? What is a communicable disease?
2. Define source, reservoir, period of infectivity, and carrier.
3. What types of infectious disease carriers does epidemiology recognize?
4. Describe the four main types of infectious disease transmission and give examples of each.
5. Define the terms droplet nuclei, vehicle, fomite, and vector.

37.7 Virulence and the Mode of Transmission

There is evidence that a pathogen's virulence may be strongly influenced by its mode of transmission and ability to live outside its host. When the pathogen uses a mode of transmission such as direct contact, it cannot afford to make the host so ill that it will not be transmitted effectively. This is the case with the common cold, which is caused by rhinoviruses and several other respiratory viruses. If the virus reproduced too rapidly and damaged its host extensively, the person would be bedridden and not contact others. The efficiency of transmission would drop because rhinoviruses shed from the cold sufferer could not contact new hosts and would be inactivated by exposure. Cold sufferers must be able to move about and directly contact others. Thus virulence is low and people are not incapacitated by the common cold.

On the other hand, if a pathogen uses a mode of transmission not dependent on host health and mobility, then the person's health will not be a critical matter. The pathogen might be quite successful—that is, transmitted to many new hosts even though it kills its host relatively quickly. Host death will mean the end of any resident pathogens, but the species as a whole can spread and flourish as long as the increased transmission rate outbalances the loss due to host death. This situation may arise in several ways.

When a pathogen is transmitted by a vector, it will be benefited by extensive reproduction and spread within the host. If pathogen levels are very high in the host, a vector such as a biting insect has a better chance of picking up the pathogen and transferring it to a new host. Indeed, pathogens transmitted by biting arthropods such as mosquitoes often are very virulent (e.g., malaria, typhus, sleeping sickness). It is important that such pathogens take good care of their vectors, and the vector generally remains healthy, at least long enough for pathogen transmission.

Virulence also is often directly correlated with a pathogen's ability to survive in the external environment. If a pathogen cannot survive well outside its host and does not use a vector, it depends on host survival and will tend to be less virulent. When a pathogen can survive for long periods outside its host, it can afford to leave the host and simply wait for a new one to come along. This seems to promote increased virulence. Host health is not critical, but extensive multiplication within the host will increase the efficiency of transmission. Good examples are tuberculosis and diphtheria. *Mycobacterium* and *Corynebacterium* survive for a long time, at least weeks to months, outside human hosts.

Human cultural patterns and behavior almost certainly also affect pathogen virulence. Waterborne pathogens such as *Vibrio cholerae* (which causes diarrhea) are transmitted through drinking water systems. They can be virulent because immobile hosts still shed pathogens, which frequently reach the water. By this argument, establishment of uncontaminated drinking water supplies should reduce cholera virulence, and this seems to be the case. The same appears to be true of *Shigella* and bacillary dysentery. Often one of the best ways to reduce virulence may be to reduce the frequency of transmission.

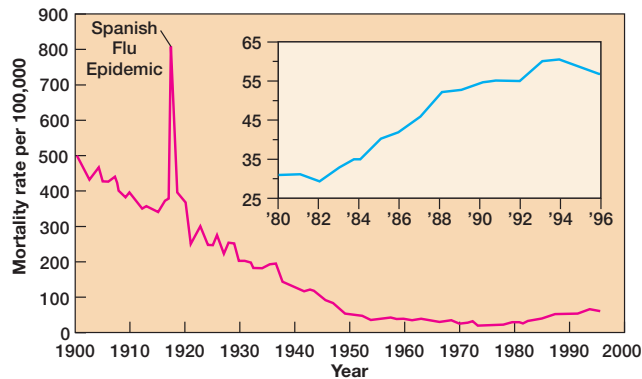


Figure 37.7 Infectious Disease Mortality in the United States Decreased Greatly during Most of the Twentieth Century. The insert is an enlargement of the right-hand portion of the graph and shows that the death rate from infectious diseases increased between 1980 and 1994.

37.8 Emerging and Reemerging Infectious Diseases and Pathogens

Only a few decades ago, a grateful public trusted that science had triumphed over infectious diseases by building a fortress of health protection. Antibiotics, vaccines, and aggressive public health campaigns had yielded a string of victories over old enemies like whooping cough, pneumonia, polio, and smallpox. In developed countries, people were lulled into believing that microbial threats were a thing of the past. Trends in the number of deaths caused by infectious diseases in the United States from 1900 through 1982 supported this conclusion (**figure 37.7**). However, this downward trend ended in 1982 and the death rate has risen in the past 20 years. In these 20 years, the world has seen the global spread of AIDS, the resurgence of tuberculosis, and the appearance of new enemies like hantavirus pulmonary syndrome, hepatitis C and E, Ebola virus, Lyme disease, cryptosporidiosis, and the deadly *E. coli* O157:H7. In addition, during this same time period:

- A “bird flu” virus that had never before attacked humans began to kill people in Hong Kong.
- A new variant of a fatal brain disease, Creutzfeldt-Jakob disease (see section 38.5 and table 38.6), was identified in the United Kingdom, apparently transmitted by beef from animals with “mad cow disease.”
- *Staphylococcus* bacteria with increased resistance to vancomycin, long the antibiotic of first choice, were seen for the first time.
- The United States was hit with several major multistate foodborne outbreaks, including those caused by parasites

on raspberries, viruses on strawberries, and bacteria in produce, ground beef, cold cuts, and breakfast cereal.

- A new strain of tuberculosis that is resistant to many drugs, and occurs most often in people infected with HIV, arose in the city of New York and other large cities.

By the 1990s, the idea that infectious diseases no longer posed a serious threat to human health was obsolete. It is now clear that globally, humans will continually be faced with both new infectious diseases and the reemergence of older diseases once thought to be conquered (e.g., tuberculosis, dengue hemorrhagic fever, yellow fever). William H. McNeill in *Plagues and Peoples* (1976) addresses this problem as follows: “Ingenuity, knowledge, and organization alter but cannot cancel humanity’s vulnerability to invasion by parasitic forms of life. Infectious disease which antedated the emergence of humankind will last as long as humanity itself, and will surely remain, as it has been hitherto, one of the fundamental parameters and determinants of human history.”

The Centers for Disease Control and Prevention (CDC) has defined these diseases as “new, reemerging, or drug-resistant infections whose incidence in humans has increased within the past two decades or whose incidence threatens to increase in the near future.” Some of the most recent examples of these diseases are shown in **figure 37.8**. The increased importance of emerging and reemerging infectious diseases has stimulated the establishment of a field called **systematic epidemiology**, which focuses on the ecological and social factors that influence the development of these diseases.

After a century marked by dramatic advances in medical research and drug discovery, technology development, and sanitation, why are viruses, bacteria, fungi, and parasites posing such a problem and challenge? Many factors characteristic of the modern world undoubtedly favor the development and spread of these microorganisms and their diseases. Examples include:

1. Unprecedented worldwide population growth, population shifts (demographics), and urbanization
2. Increased international travel
3. Increased worldwide transport (commerce), migration, and relocation of animals and food products
4. Changes in food processing, handling, and agricultural practices
5. Changes in human behavior, technology, and industry
6. Human encroachment on wilderness habitats that are reservoirs for insects and animals that harbor infectious agents
7. Microbial evolution (e.g., selection pressure) and the development of resistance to antibiotics and other antimicrobial drugs (e.g., penicillin-resistant *Streptococcus pneumoniae*, methicillin-resistant *Staphylococcus aureus*, and vancomycin-resistant enterococci)
8. Changes in ecology and climate
9. Modern medicine (e.g., immunosuppression)

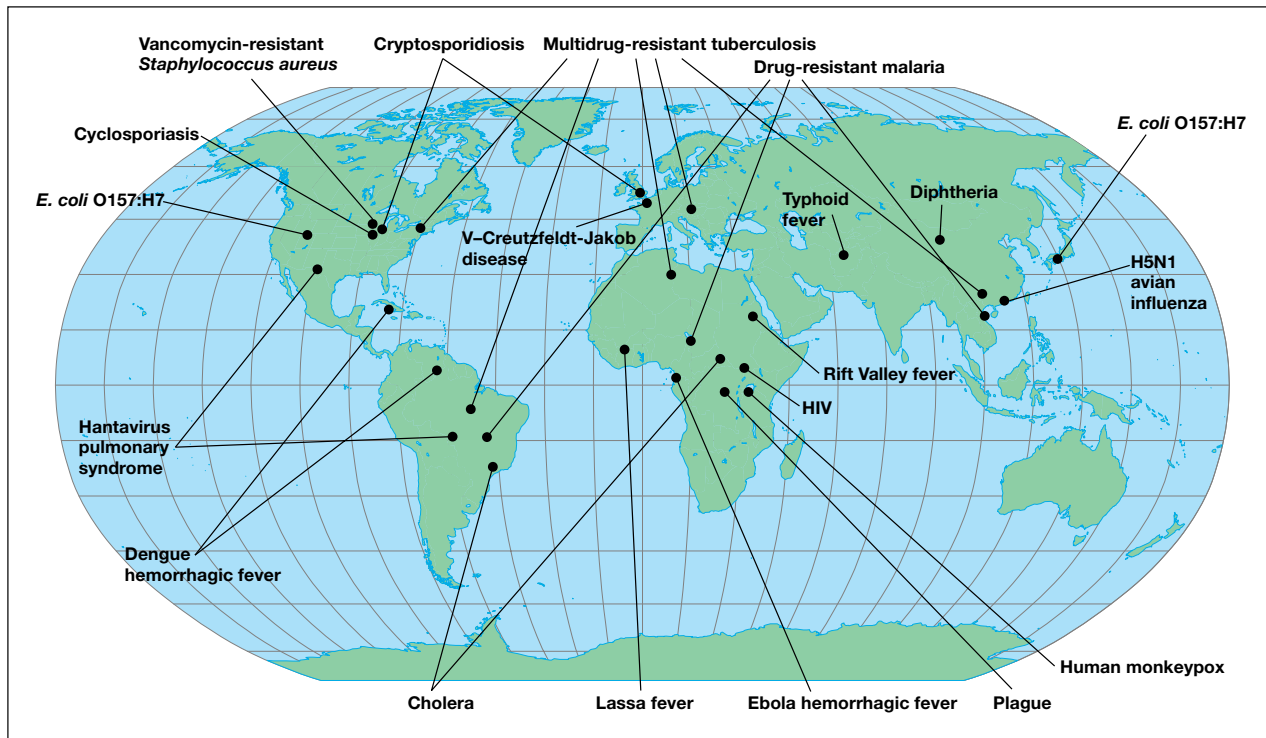


Figure 37.8 Some Examples of Emerging and Reemerging Infectious Diseases. Although diseases such as HIV are indicated in only one or two significant locations, they are very widespread and a threat in many regions. See text for further details.

10. Inadequacy of public infrastructure and vaccination programs
11. Social unrest and civil wars
12. The possibility of bioterrorism (section 37.10)
13. Virulence-enhancing mechanisms of pathogens (e.g., the mobile genetic elements—bacteriophages, plasmids, transposons)

A more detailed discussion of some of these examples now follows.

Reasons for Increases in Emerging and Reemerging Infectious Diseases

As just noted, many factors contribute to increases in infectious disease. Economic or military forces that cause population shifts create fertile ground for diseases to emerge and spread. Changes in human behavior, whether in sexual practices, the use of IV drugs, or food preferences, affect microbial spread.

Food practices that include handling, cutting, refrigeration, and other treatments to maintain quality also disseminate pathogenic microorganisms and select for those that grow under these conditions. The globalization of food processing and supply centers has created the potential for widespread outbreaks of food-borne microbial diseases originating from a single batch of

tainted processed foods or contaminated raw foods, mishandling of food shipments, or distribution of microbe-laden fresh produce. The CDC estimates that food-borne diseases cause approximately 76 million illnesses, 350,000 hospitalizations, and 5,000 deaths each year in the United States alone. Known pathogens are responsible for only 18% of these illnesses, and *Listeria*, *Salmonella*, and *Toxoplasma* are responsible for 75% of food-related deaths globally. Among the incidences in recent years are the following: (1) the 1994 outbreak of salmonellosis due to contaminated ice cream; (2) the 1996 and 1997 outbreaks of *Cyclospora* infections linked to Guatemalan raspberries; and (3) the periodic outbreaks in recent years of *E. coli* O157:H7 related to consumption of raw or undercooked beef and unpasteurized apple juice (it should be noted that this pathogen causes hemolytic uremic syndrome in children; see section 39.4).

As population density increases in cities, the dynamics of microbial exposure and evolution increase in humans themselves. Urbanization often crowds humans and increases exposure to microorganisms. Crowding leads to unsanitary conditions and hinders the effective implementation of adequate medical care, enabling more widespread transmission and propagation of pathogens. In modern societies, crowded workplaces, community-living settings, day-care centers, large hospitals, and public transportation all facilitate microbial transmission. Furthermore,

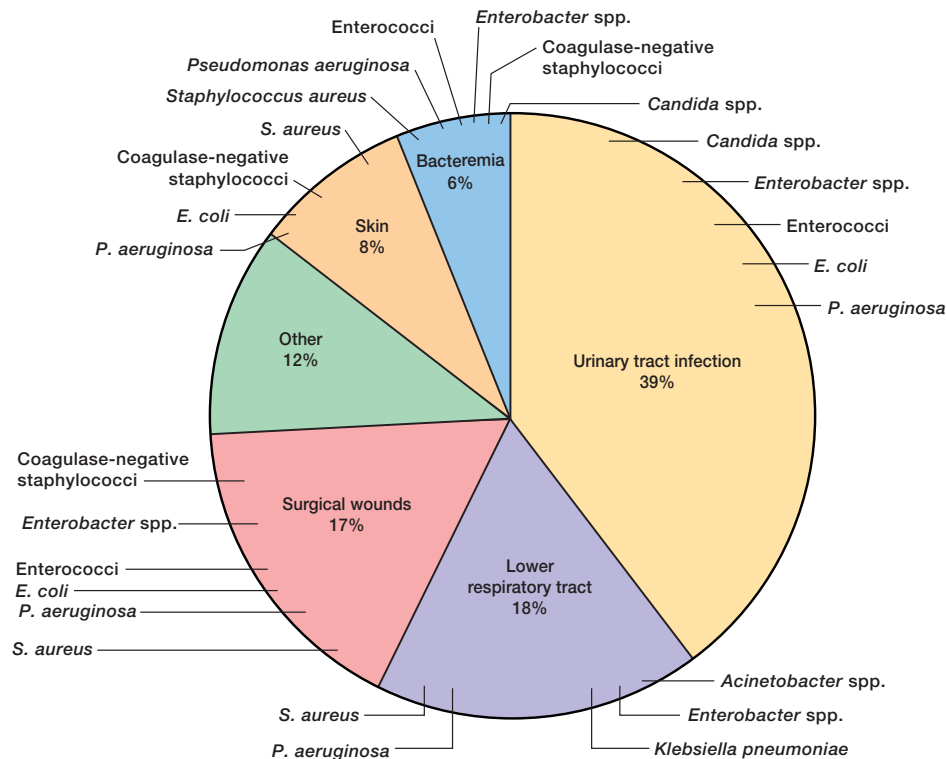


Figure 37.9 Nosocomial Infections. Relative frequency by body site. These data are from the National Nosocomial Infections Surveillance, which is conducted by the Centers for Disease Control and Prevention (CDC).

land development and the exploration and destruction of natural habitats have increased the likelihood of human exposure to new pathogens and may put selective pressures on pathogens to adapt to new hosts and changing environments. The introduction of pathogens to a new environment or host can alter transmission and exposure patterns, leading to sudden proliferation of disease. For example, the spread of Lyme disease in New England probably was due partly to ecological disruption that eliminated predators of deer. An increase in deer and the deer tick populations provided a favorable situation for pathogen spread to humans. Whenever there is alteration of the environment and new environments are created, this may not only confer a survival advantage, but may also increase a pathogen's virulence and alter its drug susceptibility profile. When there are changes in climate or ecology, it should not be surprising to find changes in both beneficial and detrimental microorganisms. Global warming also affects microorganism selection and survival. Finally, mass migrations of refugees, workers, and displaced persons have led to a steady growth of urban centers at the expense of rural areas.

Microbiologists are all too familiar with the development of resistance to antibiotics used in human medicine. The distribution of nosocomial pathogens (figure 37.9) has changed throughout the antibiotic era. Hospital acquired infections were dominated early on by staphylococci, which initially responded to penicillin. During subsequent years, the emergence of methicillin-resistant *Staphylococcus aureus* (MRSA) increased from 2% in 1975, to

14% in 1987, and to over 40% in 1999; similar patterns are emerging for penicillin-resistant *Streptococcus pneumoniae*. *S. aureus* with intermediate resistance to vancomycin began to appear in 1997 and the CDC believes that glycopeptide-resistant *S. aureus* infections are inevitable. Glycopeptide-resistant *Enterococcus faecium* was first reported in the late 1980s; in 1993 nearly 14% of enterococci-associated nosocomial infections were caused by vancomycin-resistant enterococci and that number rose to 25% in 1998. In recent years, gram-positive cocci have reemerged as significant nosocomial pathogens. Newly recognized nosocomial, gram-positive species include *Corynebacterium jeikeium* and *Rhodococcus equi*. The incidences of infections by the gram-negative pathogens *P. aeruginosa* and *Acinetobacter* have increased. With respect to the extended-spectrum β -lactamase-resistant gram-negative bacilli, bacteria such as *Klebsiella pneumoniae*, *Escherichia coli*, other *Klebsiella* spp., *Proteus* spp., *Morganella* spp., *Citrobacter* spp., *Salmonella* spp., and *Serratia marcescens* are resistant to penicillins; first-generation cephalosporins; and some third-generation cephalosporins such as cefotaxime (Claforan), ceftriaxone (Rocephin), ceftazidime, and aztreonam (Azactam). Newly renamed gram-negative bacterial pathogens include *Burkholderia cepacia* and *Stenotrophomonas maltophilia*.

Without a doubt, the key factors responsible for the rise in drug-resistant pathogens have been the excessive or inappropriate use of antimicrobial therapy and the sometimes indiscriminant

use of broad-spectrum antibiotics. Although the CDC acknowledges that it is too late to solve the resistance problem simply by using antimicrobial agents more prudently, it is no less true that the problem of drug resistance will continue to worsen if this does not occur. Also needed (especially in underdeveloped countries) is a renewed emphasis on alternate prevention and control strategies that prevailed in the years before antimicrobial chemotherapy. These include improved sanitation and hygiene, isolation of infected persons, antisepsis, and vaccination. [Drug resistance \(pp. 818–20\).](#)

One of the paradoxes of modern medicine is that while we are managing and even eradicating some infectious diseases, others are emerging. Immunosuppression, either by another disease agent such as the AIDS virus or by drugs taken upon organ transplantation, increases the number of individuals susceptible to new pathogens. Immunocompromised patients represent an expanding intermediate habitat, providing the opportunity (not available in immunocompetent patients) for a pathogen to adapt to the human host and acquire the capability to circumvent the immune system of a healthy person. These immunocompromised populations can then serve as a reservoir for emerging and reemerging infections, which may eventually spread beyond the confines of institutional settings to the community at large.

In this new millennium, the speed and volume of international travel are major factors contributing to the global emergence of infectious diseases. The spread of a new disease often used to be limited by the travel time needed to reach a new host population. If the travel time was sufficiently long, as when a ship crossed the ocean, the infected travelers would either recover or die before reaching a new population. Because travel by air has obliterated time between exposure and disease outbreak, a traveler can spread virtually any human disease in a matter of hours. Vehicles of human transport, such as aircraft and ships, also transport the infectious agents and their vectors.

It is probably best to view emerging and reemerging pathogens and their diseases as the outcome of many different factors. Because the world is now so interrelated, we cannot isolate ourselves from other countries and continents. Changes in the disease status of one part of the world may well affect the health of the remainder. As Nobel laureate Joshua Lederberg has so eloquently stated, “The microbe that felled one child in a distant continent yesterday can reach your child today and seed a global pandemic tomorrow.”

1. Describe how virulence and the mode of transmission may be related. What might cause the development of new human diseases?
2. How would you define emerging or reemerging infectious diseases?
3. What are some of the factors responsible for the emergence or reemergence of pathogens?
4. What are some key factors responsible for the rise in drug-resistant bacteria?

37.9 Control of Epidemics

The development of an infectious disease is a complex process involving many factors, as is the design of specific epidemiological control measures. Epidemiologists must consider available resources and time constraints, adverse effects of potential control measures, and human activities that might influence the spread of the infection. Many times control activities reflect compromises among alternatives. To proceed intelligently, one must identify components of the infectious disease cycle that are primarily responsible for a particular epidemic. Control measures should be directed toward that part of the cycle that is most susceptible to control—the weakest link in the chain.

There are three types of control measures. The first type is directed toward reducing or eliminating the source or reservoir of infection:

1. Quarantine and isolation of cases and/or carriers
2. Destruction of an animal reservoir of infection
3. Treatment of sewage to reduce water contamination
4. Therapy that reduces or eliminates infectivity of the individual

The second type of control measure is designed to break the connection between the source of the infection and susceptible individuals. Examples include general sanitation measures:

1. Chlorination of water supplies
2. Pasteurization of milk
3. Supervision and inspection of food and food handlers
4. Destruction of vectors by spraying with insecticides

The third type of control measure reduces the number of susceptible individuals and raises the general level of herd immunity by immunization. Examples include the following:

1. Passive immunization to give a temporary immunity following exposure to a pathogen or when a disease threatens to take an epidemic form
2. Active immunization to protect the individual from the pathogen and the host population from the epidemic

The Role of the Public Health System: Epidemiological Guardian

The control of an infectious disease relies heavily on a well-defined network of clinical microbiologists, nurses, physicians, epidemiologists, and infection control personnel who supply epidemiological information to a network of local, state, national, and international organizations. These individuals and organizations comprise the public health system. For example, each state has a public health laboratory that is involved in infection surveillance and control. The communicable disease section of a state laboratory includes specialized laboratory services for the examination of specimens or cultures submitted by physicians, the local health department, hospital laboratories, sanitarians, epidemiologists, and others. These groups share their findings with other health agencies in the state, the Centers for Disease Control and Prevention, and the World Health Organization (WHO).

37.10 The Emerging Threat of Bioterrorism

Today, **bioterrorism** (Greek *bios*, life, and terrorism, the systematic use of terror to demoralize, intimidate, and subjugate) is a reality. The Centers for Disease Control and Prevention recently defined bioterrorism as “the intentional or threatened use of viruses, bacteria, fungi, or toxins from living organisms to produce death or disease in humans, animals, and plants.” At the end of this past century, Iraq used chemical weapons on Iran and on its own citizens, and concealed a biological weapons program. During this same time period, the hitherto-unknown Japanese cult Aum Shin-rikyo used sarin nerve gas in a Tokyo subway. Currently terrorist incidents and hoaxes involving toxic or infectious agents have been on the rise. Somewhere, sometime in the future, terrorists may well threaten to use, or attempt to use, a biological weapon against the United States or another country. Although the arsenal of the terrorist is potentially large, this discussion of bioterrorism focuses only on the biological threats.

Among weapons of mass destruction, biological weapons are more destructive than chemical weapons, including nerve gas. In certain circumstances, biological weapons can be as devastating as a nuclear explosion—a few kilograms of anthrax can kill as many people as a Hiroshima-size nuclear bomb. In 1998, the U.S. government launched the first national effort to create a biological weapons defense. The initiatives include (1) the first-ever procurement of specialized vaccines and medicines for a national civilian protection stockpile; (2) invigoration of research and development in the science of biodefense; (3) investment of more time and money in genome sequencing, new vaccine research, and new therapeutic research; (4) development of improved detection and diagnostic systems; and (5) preparation of clinical microbiologists and the clinical microbiology laboratory as members of the “first responder” team, which is to respond in a timely manner to acts of bioterrorism.

The list of biological agents that could pose the greatest public health risk in the event of a bioterrorist attack is short and includes viruses, bacteria, parasites, and toxins (**table 37.3**). Although short, the list includes agents that, if acquired and properly disseminated, could become a difficult public health challenge in terms of limiting the numbers of casualties and controlling damage.

As noted in section 33.1, vaccination has been the single most cost-effective public health intervention that has shielded populations from a dozen serious and sometimes fatal naturally transmitted illnesses. In evaluating the role of vaccines for protecting civilian populations from bioterrorism, new problems arise. Despite the protective efficacy of vaccines against individual microorganisms, the very high costs and the great difficulties involved in vaccinating large populations, along with the broad spectrum of potential agents, make it impossible to use vaccines to protect a population against bioterrorism.

In the United States, the U.S. Food and Drug Administration has been charged to foster the development of vaccines, drugs and diagnostic products, food supply safeguards, and other measures needed to respond to bioterrorist threats (**table 37.4**). The expected outcomes of these activities include safe and effective products to treat or prevent toxicity of biological and chemical

Table 37.3 Biological Agents Associated with Bioterrorism or Biocrimes

	Traditional Biological Warfare Agents	Agents of Biocrimes and Bioterrorism
Pathogens	Smallpox virus Viral encephalitides Viral hemorrhagic fevers <i>Bacillus anthracis</i> <i>Brucella suis</i> <i>Coxiella burnetii</i> <i>Francisella tularensis</i> <i>Yersinia pestis</i>	<i>Ascaris suum</i> <i>Bacillus anthracis</i> <i>Coxiella burnetii</i> <i>Giardia lamblia</i> HIV <i>Rickettsia prowazekii</i> (typhus) <i>Salmonella typhi</i> <i>Schistosoma</i> spp. <i>Vibrio cholerae</i> Viral hemorrhagic fevers (Ebola) Yellow fever virus <i>Yersinia enterocolitica</i> <i>Yersinia pestis</i> (plague)
Toxins	Botulinum Ricin Staphylococcal enterotoxin B	Botulinum Cholera endotoxin Diphtheria toxin Nicotine Ricin Snake toxin Tetrodotoxin
Anticrop agents	Rice blast Rice stem rust Wheat stem rust	

Source: Adapted from the 2000 NATO Handbook on the Medical Aspects of NBC Defensive Operations.

agents; methods to rapidly detect, identify, and decontaminate hazardous microorganisms; a greater ability to ensure the safety of the food supply; and greater capacity to provide appropriate medical care and a public health response.

1. In what three general ways can epidemics be controlled? Give one or two specific examples of each type of control measure.
2. Name some of the microorganisms that can be used to commit biocrimes. From this list, which two pose the greatest risk for causing large numbers of casualties?
3. Why are biological weapons more destructive than chemical weapons?
4. What kinds of things can be done to defend a population against bioterrorism?

37.11 Global Travel and Health Considerations

From a global health perspective, developed countries such as Australia, the European countries, Israel, New Zealand, and the United States have highly effective public health systems. Approximately 25% of the nearly 6 billion people on our planet

Table 37.4 Activities of the U.S. Food and Drug Administration to Counter Bioterrorism

1. Enhancing the expeditious development and licensure of new vaccines and biological therapeutics through research and review activities—anthrax vaccine and antisera to botulinum toxin, for example
2. Enhancing the timeliness of application reviews of new drugs and biological products and new and existing products
3. Participation in the planning and coordination of public health and medical responses to a terrorist attack involving biological or chemical agent(s)
4. Participation in the development of rapid detection and decontamination for agents of bioterrorism such as *Clostridium botulinum* toxins, *Yersinia pestis*, and *Bacillus anthracis*
5. Ensuring the safety of regulated foods, drugs, medical devices, and biological products; arranging for seizure and disposal of affected products
6. Developing techniques for detection of genetic modifications of microorganisms to make them more toxic or more antibiotic- or vaccine-resistant
7. Rapidly determining a microorganism's sensitivity to drug therapies
8. Determining the mechanism of replication and pathogenicity or virulence of identified microorganisms, including elements that can be transferred to other organisms to circumvent detection, prevention, or treatment
9. Enhancing bioterrorism agent reporting and surveillance capabilities

Earth live in these countries. As a result, of the approximately 12 million deaths in these countries per year, only about 500,000 are due to infectious diseases. Underdeveloped areas such as Africa, Central and South America, India, the New Independent States of the former Soviet Union, and Asia do not have highly developed public health systems and represent the other 75% of the human population. It is in these regions that infectious diseases are the major cause of death; for example, of approximately 38.5 million deaths per year, about 18 million are attributed to infectious microbial diseases.

The high incidence of infectious microbial diseases in underdeveloped countries must be of great concern for people traveling to these destinations. Each year 1 billion passengers travel by air, and over 50 million people from developed countries visit underdeveloped countries. Furthermore, the time required to circumnavigate the globe has decreased from 365 days to fewer than 3 days.

Several kinds of precautions can be taken by individuals to prevent travel-related infectious diseases. Examples include:

1. If one is traveling to an area where malaria is endemic, weekly prophylaxis before entering the area and after leaving the area is recommended.
2. When traveling, individuals often have a sense of anonymity; they may feel less inhibited sexually and place themselves at greater risk of acquiring sexually transmitted diseases. Travelers should recall the benefits of abstinence or protective sexual practices, especially the use of condoms. Hepatitis B vaccine should be administered if it is indicated.
3. Travelers should avoid uncooked food, nonbottled water and beverages, and unpasteurized dairy products. Use bottled water for drinking, making ice cubes, and brushing teeth.
4. Wash hands with soap and water frequently, especially before each meal.
5. To prevent respiratory infections, avoid excessive outdoor activities in areas of heavy air pollution during hot or humid parts of the day. Consider tuberculin skin testing before and after travel.

6. Minimize skin exposure and use repellents to prevent arthropod-borne illnesses (e.g., malaria, dengue, yellow fever, Japanese encephalitis).
7. Avoid skin-perforating procedures (e.g., acupuncture, body piercing, tattooing, venipuncture, sharing of razors).
8. Do not pet or feed animals, especially dogs and monkeys.
9. Avoid swimming or wading in nonchlorinated freshwater.

Vaccinations are one of the most important strategies of prophylaxis in travel medicine. A medical consultation before travel is an excellent opportunity to update routine immunizations. Selection of immunizations should be based on requirements and risk of infection at the travel destination. According to International Health Regulations, many countries require proof of yellow fever vaccination on the International Certificate of Vaccination. Additionally, a few countries still require proof of vaccination against cholera, diphtheria, and meningococcal disease. The basic CDC immunization recommendations for those traveling abroad are presented in **table 37.5**. In the not-too-distant future, travelers will be offered a variety of oral vaccines against the microorganisms causing dengue fever and travelers' diarrhea (e.g., enterotoxigenic *Escherichia coli*, *Campylobacter*, *Shigella*); vaccines against malaria and AIDS, the infections causing most deaths in travelers, are much farther in the future.

In addition, a traveler should:

1. Read carefully CDC information about the destination and follow recommendations
2. Begin the vaccination process early
3. Find a travel clinic for information and specialized immunizations
4. Plan ahead when traveling with children or if there are special needs
5. Learn about safe food and water (contaminated food and water are the major sources of stomach or intestinal illness while traveling), protection against insects, and other precautions
6. Prepare for medical emergencies and for nonmedical emergencies such as crime and natural disasters

Table 37.5 Vaccine Recommendations for Travelers Aged 2 Years or Older**For Travelers Aged 2 Years or Older, the Following Immunizations Normally Given During Childhood (see figure 33.1) Should Be Up-to-Date:**

1. Measles, mumps, and rubella (MMR) vaccine—at least 1 dose given on or after 12 months of age
2. Diphtheria, tetanus, and acellular pertussis (DtaP)—4 or 5 doses until age 7; after age 7, 1 dose of adult tetanus and diphtheria (Td) vaccine every 10 years
3. Polio vaccine—at least 3 doses
4. *Haemophilus influenzae* type b (Hib) vaccine—not recommended after 5 years of age
5. Hepatitis B vaccine—3 doses
6. Varicella vaccine (for persons who have never had chickenpox)

In Addition, Depending on the Country, Travelers Should Consider:

1. Influenza (flu) vaccine—recommended for adults 65 years or older and for other high-risk individuals
2. Pneumococcal vaccine—recommended for adults 65 years or older and for other high-risk individuals

Booster or Additional Doses:

1. A booster dose of adult tetanus-diphtheria (Td) recommended every 10 years
2. For persons who have received a complete series of polio vaccine, an additional single dose of inactivated polio vaccine should be given to persons over 18 traveling to the following areas: Africa, Asia, the Middle East, India, and the New Independent States of the former Soviet Union
3. Persons born after 1957 should have a second dose of measles vaccine before traveling abroad

The Following Immunizations Are Recommended Based on Destination:

<i>Immunization</i>	<i>Destination</i>
Yellow fever	Africa and South America
Hepatitis A or immune globulin (IG)	To all areas except Japan, Australia, New Zealand, northern and western Europe, and North America (excluding Mexico)
Hepatitis B (if staying 6 or more months)	Southeast Asia, Africa, the Middle East, the islands of South and Western Pacific, and the Amazon region of South America
Typhoid	In all developing countries and for those travelers spending time in areas where food and water precautions are recommended
Meningococcus	Sub-Saharan Africa
Japanese encephalitis or tick-borne encephalitis	Areas of risk
Cholera	The risk of cholera to U.S. travelers is so low that it is questionable whether cholera vaccine is of benefit

Spacing of Immunobiologics

All vaccines (except cholera and yellow fever) may be safely administered simultaneously without any decrease in effectiveness. Immune globulin (IG) may be simultaneously administered at different body locations with an inactivated vaccine such as DtaP, IPV, Hib, and hepatitis A and B vaccines. However, IG diminishes the effectiveness of live virus MMR and varicella vaccines if IG is given simultaneously. IG does not interfere with either oral polio vaccine (OPV) or yellow fever vaccine when given simultaneously.

Women who are pregnant or who are likely to become pregnant within 3 months should not receive MMR or varicella vaccines. Yellow fever vaccine or OPV should be given to pregnant women only if there is a substantial risk of exposure.

Source: National Center for Infectious Diseases, *Travelers' Health*, CDC Vaccine Recommendations (Year 2000).

Space Travel

Space travel entails unique circumstances that influence the health of astronauts, most notably the limitations imposed by zero-gravity environments. The impact of weightlessness on human physiology is substantial and affects various body systems: neurologic, psychologic, vestibular, cardiovascular, musculoskeletal, endocrine, hematologic, and immunologic. For example, recent studies on the immune system have documented a decrease in cell-mediated immunity. The space environment also challenges the growth of microorganisms. Increased antimicrobial resistance has been noted among *E. coli* and *Staphylococcus aureus* bacteria in space, and fungal overgrowths can be a problem because of changes in humidity and pockets of increased condensation aboard spacecraft and space stations.

The primary strategies used by NASA to prevent health-related mishaps associated with space travel include selecting exceptionally fit, healthy crew members and imposing preflight protective quarantines. Access to astronauts is restricted to family members beginning 10 days before departure and a strict 7-day pre-launch quarantine is imposed.

1. Give some examples where population movements affect microbial disease transmission.
2. In addition to vaccinations, what are some additional precautions global travelers should take?
3. What are some of the health problems associated with space travel?

37.12 Nosocomial Infections

Nosocomial infections [Greek *nosos*, disease, and *komeion*, to take care of] result from pathogens that develop within a hospital or other type of clinical care facility and are acquired by patients while they are in the facility. Besides harming patients, nosocomial infections can affect nurses, physicians, aides, visitors, salespeople, delivery personnel, custodians, and anyone who has contact with the hospital. Most nosocomial infections become clinically apparent while patients are still hospitalized; however, disease onset can occur after patients have been discharged. Infections that are incubating when patients are admitted to a hospital are not nosocomial; they are community acquired. However, because such infections can serve as a ready source or reservoir of pathogens for other patients or personnel, they are also considered in the total epidemiology of nosocomial infections.

The Centers for Disease Control and Prevention estimate that 5 to 10% of all hospital patients acquire some type of nosocomial infection. Because approximately 40 million people are admitted to hospitals annually, about 2 to 4 million people may develop an infection they did not have upon entering the hospital. Thus nosocomial infections represent a significant proportion of all infectious diseases acquired by humans.

Nosocomial diseases are usually caused by bacteria, most of which are noninvasive and part of the normal microbiota; viruses, protozoa, and fungi are rarely involved. Figure 37.9 summarizes the most common types of nosocomial infections and the most common nosocomial pathogens.

Source

The nosocomial pathogens that cause diseases come from either endogenous or exogenous sources. Endogenous sources are the patient's own microbiota; exogenous sources are microbiota other than the patient's. Endogenous pathogens are either brought into the hospital by the patient or are acquired when the patient becomes colonized after admission. In either case the pathogen colonizing the patient may subsequently cause a nosocomial disease (e.g., when the pathogen is transported to another part of the body or when the host's resistance drops). If it cannot be determined that the specific pathogen responsible for a nosocomial disease is exogenous or endogenous, then the term **autogenous** is used. An **autogenous infection** is one that is caused by an agent derived from the microbiota of the patient, despite whether it became part of the patient's microbiota following his or her admission to the hospital.

There are many potential exogenous sources in a hospital. Animate sources are the hospital staff, other patients, and visitors. Some examples of inanimate exogenous sources are food, urinary catheters, intravenous and respiratory therapy equipment, and water systems (softeners, dialysis units, and hydrotherapy equipment).

Control, Prevention, and Surveillance

In the United States nosocomial infections prolong hospital stays by 4 to 13 days, result in over 4.5 billion dollars a year in direct hospital charges, and lead to over 20,000 direct and 60,000 indirect deaths annually. The enormity of this problem has led most hospitals to allocate substantial resources to the development of methods and programs for the surveillance, prevention, and control of nosocomial infections.

All personnel involved in the care of patients should be familiar with basic infection control measures such as isolation policies of the hospital; aseptic techniques; proper handling of equipment, supplies, food, and excreta; and surgical wound care and dressings. To adequately protect their patients, hospital personnel must practice proper aseptic technique and hand-washing procedures, and must wear gloves when contacting mucous membranes, secretions, and "moist body substances." Patients should be monitored with respect to the frequency, distribution, symptomatology, and other characteristics common to nosocomial infections. A dynamic control and surveillance program can be invaluable in preventing many nosocomial infections, patient discomfort, extended stays, and further expense.

The Hospital Epidemiologist

Because of nosocomial infections, all hospitals desiring accreditation by the Joint Commission on Accreditation of Healthcare Organizations (JCAHO) must have a designated individual directly responsible for developing and implementing policies governing control of infections and communicable diseases. This individual may be a registered nurse and is known as a hospital epidemiologist, nurse epidemiologist, infection control nurse, infection control practitioner, or a clinical microbiologist/technologist. In larger hospitals, a physician is the hospital epidemiologist and should be trained in infectious diseases. He or she oversees a staff that includes nurse epidemiologists, quality assurance specialists, fellows in infectious disease/hospital epidemiology, and epidemiology technicians. The hospital epidemiologist must meet with an infection control committee composed of various professionals who have expertise in the different aspects of infection control and hospital operation. The infection control committee periodically evaluates laboratory reports, patients' charts, and surveys done by the hospital epidemiologist to determine whether there has been any increase in the frequency of particular infectious diseases or potential pathogens.

Overall, the services provided by the hospital epidemiologist should include at least the following:

1. Research in infection control
2. Evaluation of disinfectants, rapid test systems, and other products
3. Efforts to encourage appropriate legislation related to infection control, particularly at the state level

4. Efforts to contain hospital operating costs, especially those related to fixed expenses such as the DRGs (diagnosis-related groups)
5. Surveillance and comparison of endemic and epidemic infection frequencies
6. Direct participation in a variety of hospital activities relating to infection control and maintenance of employee health
7. Education of hospital personnel in communicable disease control and disinfection and sterilization procedures
8. Establishment and maintenance of a system for identifying, reporting, investigating, and controlling infections and communicable diseases of patients and hospital personnel
9. Maintenance of a log of incidents related to infections and communicable diseases

10. Monitoring trends in the antimicrobial drug resistance of infectious agents

Recently computer software packages have been developed to aid the infection control practitioner. Such packages generate standard reports, cause-and-effect tabulations, and graphics for the daily epidemiological monitoring that must be done.

-
1. Describe a nosocomial infection.
 2. What two general sources are responsible for nosocomial infections? Give some specific examples of each general source.
 3. Why are nosocomial infections important?
 4. What does a hospital epidemiologist do to control nosocomial infections?
-

Summary

1. Epidemiology is the science that evaluates the determinants, occurrence, distribution, and control of health and disease in a defined population.
2. Specific epidemiological terminology is used to communicate disease incidence in a given population. Frequently used terms include sporadic disease, endemic disease, hyperendemic disease, epidemic, index case, outbreak, and pandemic.
3. An important tool used in the study of modern epidemiology is statistics.
4. Epidemiological data can be obtained from such factors as morbidity, prevalence, and mortality rates.
5. An infectious disease is caused by microbial agents such as viruses, bacteria, fungi, protozoa, and helminths that can be transmitted from one host to another.
6. The manifestations of an infectious disease can range from mild to severe to deadly, depending on the agent and host.
7. Surveillance is necessary for recognizing a specific infectious disease within a given population. This consists of gathering data on the occurrence of the disease, collating and analyzing the data, summarizing the findings, and applying the information to control measures.
8. Remote sensing and geographic information systems can be used to gather epidemiological data on the environment.
9. A common-source epidemic is characterized by a sharp rise to a peak and then a rapid, but not as pronounced, decline in the number of individuals infected (**figure 37.2**). A propagated epidemic is characterized by a relatively slow and prolonged rise and then a gradual decline in the number of individuals infected.
10. Herd immunity is the resistance of a population to infection and pathogen spread because of the immunity of a large percentage of the individuals within the population (**figure 37.4**).
11. The infectious disease cycle or chain involves the characteristics of the pathogen, the source and/or reservoir of the pathogen, the transmission of the pathogen, the susceptibility of the host, the exit mechanism of the pathogen from the body of the host, and its spread to a new reservoir or host (**figure 37.5**).
12. There are four major modes of transmission: airborne, contact, vehicle, and vector-borne.
13. The degree of virulence may be influenced by the pathogen's preferred mode of transmission. New human diseases may arise and spread because of ecosystem disruption, rapid transportation, human behavior, and other factors.
14. It is now clear that globally, humans will continually be faced with both new infectious diseases and the reemergence of older diseases once thought to be conquered.
15. CDC has defined these diseases as "new, reemerging, or drug-resistant infections whose incidence in humans has increased within the past two decades or whose incidence threatens to increase in the near future" (**figure 37.8**).
16. Many factors characteristic of the modern world undoubtedly favor the development and spread of these microorganisms and their diseases.
17. The public health system consists of individuals and organizations that function in the control of infectious diseases and epidemics.
18. Epidemiological control measures can be directed toward reducing or eliminating infection sources, breaking the connection between sources and susceptible individuals, or isolating the susceptible individuals and raising the general level of herd immunity by immunization.
19. Today bioterrorism is a reality. Terrorist incidents and hoaxes involving toxic or infectious agents have been on the rise.
20. Among weapons of mass destruction, biological weapons are more destructive than chemical weapons. The list of biological agents that could pose the greatest public health risk in the event of a bioterrorist attack is short and includes viruses, bacteria, parasites, and toxins (**table 37.3**).
21. Certain precautions and health considerations should be taken into consideration when traveling globally.
22. Vaccinations are one of the most important strategies of prophylaxis in travel medicine (**table 37.5**).
23. Space travel entails unique considerations that influence the health of astronauts, most notably limitations imposed by zero-gravity environments. The space environment also challenges the growth of microorganisms.
24. Nosocomial infections are infections acquired during hospitalization and are produced by a pathogen acquired during a patient's stay. These infections come from either endogenous or exogenous sources (**figure 37.9**).
25. Hospitals must designate an individual to be responsible for identifying and controlling nosocomial infections. This person is known as a hospital epidemiologist, nurse epidemiologist, infection control nurse, or infection control practitioner.

Key Terms

active carrier 854
acute carrier 854
airborne transmission 854
antigenic drift 852
antigenic shift 852
autogenous infection 866
biologic transmission 858
bioterrorism 863
carrier 854
casual carrier 854
chronic carrier 854
common-source epidemic 851
common vehicle transmission 857
communicable disease 854
contact transmission 856
convalescent carrier 854
disease 848
disease syndrome 850
droplet nuclei 856
endemic disease 849
enzootic 849

epidemic 849
epidemiologist 849
epidemiology 848
epizootic 849
epizootiology 849
fomite 857
geographic information system (GIS) 850
harborage transmission 858
health 848
healthy carrier 854
herd immunity 851
hyperendemic disease 849
incubation period 850
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remote sensing (RS) 850
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symptoms 850
systematic epidemiology 859
transient carrier 854
vector 854
vector-borne transmission 857
vehicle 857
zoonoses 849

Questions for Thought and Review

1. Why is international cooperation a necessity in the field of epidemiology?
2. Are any risks involved in attempting to eliminate a pathogen from the world?
3. Why is a knowledge of statistics important to an epidemiologist?
4. What is the role of the Centers for Disease Control and Prevention in epidemiology?
5. What common sources of infectious disease are found in your community? How can the etiologic agents of such infectious diseases spread from their source or reservoir to members of your community?
6. How could you experimentally prove the cause of an infectious disease?
7. How do epidemiologists recognize an infectious disease within a given population?
8. How could you prove that an epidemic of a given infectious disease was occurring?
9. Why is the infectious disease cycle also referred to as a chain?
10. How can epidemics be controlled?
11. Why are nosocomial infections so important to humans?
12. Why do some epidemiologists believe it is impossible to prevent all nosocomial infections?
13. What is the difference between a fomite and a vector? A reservoir and a source?
14. How can changes in herd immunity contribute to an outbreak of a disease on an island?
15. What is an index case?
16. Where in the world would you expect new human diseases to arise? Why?
17. Contrast mortality due to infectious diseases in developing and developed countries.
18. What are some of the factors that are important in the reemergence of a potential pathogen?
19. Why should we be worried about bioterrorism?
20. What precautions should you take when traveling abroad?

Critical Thinking Questions

1. College dormitories are notorious for outbreaks of flu and other infectious diseases. This is particularly prevalent during final exam weeks. Using your knowledge of the immune response and epidemiology, suggest practices that could be adopted to minimize the risks at such a critical time of the term.
2. Space exploration presents an area of concern. What sort of precautions should have been taken to prevent contamination of the Martian terrain with Earth microorganisms when the Mars Lander touched down on the planet?

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CHAPTER 38

Human Diseases Caused by Viruses



These mechanical respirators, also called iron lungs, were used to keep patients alive during the polio epidemics of the 1940s through the mid 1950s. In 1955 Jonas Salk, an American microbiologist, developed the inactivated poliovirus vaccine (IPV) and in 1961, Albert Sabin, an American virologist, developed the oral poliovirus vaccine (OPV). As a result, polio has been eradicated in the United States and iron lungs are part of our history.

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Concepts

1. Some viruses can be transmitted through the air and directly or indirectly involve the respiratory system. Most of these viruses are highly communicable and cause diseases such as chickenpox, influenza, measles, mumps, respiratory syndromes and viral pneumonia, rubella, and hantavirus pulmonary syndrome.
2. The arthropod-borne diseases are transmitted by arthropod vectors from human to human or animal to human. Examples include the various encephalitides, Colorado tick fever, West Nile encephalitis, and historically important yellow fever.
3. Some viruses are so sensitive to environmental influences that they are unable to survive for significant periods of time outside their hosts. These viruses are transmitted from host to host by direct contact and cause diseases such as AIDS, cold sores, the common cold, cytomegalovirus inclusion disease, genital herpes, human herpesvirus 6 infections, human parvovirus B19 infections, certain leukemias, infectious mononucleosis, human papillomavirus, rabies, and viral hepatitis.
4. Viruses that can be transmitted by food and water and usually either grow in or pass through the intestinal system leave the body in the feces and are acquired through the oral route. Examples of such diseases include viral gastroenteritis, hepatitis A and E, and poliomyelitis.
5. The slow virus diseases represent progressive pathological processes caused by viruses or prions that remain clinically silent during a prolonged period of months or years, after which progressive clinical disease becomes apparent, usually ending months later in profound disability or death. Examples include new variant Creutzfeldt-Jakob disease, kuru, progressive multifocal leukoencephalopathy, Gerstmann-Sträussler-Scheinker syndrome, and subacute sclerosing panencephalitis.
6. One other disease associated with viruses but that does not fit into any of the foregoing categories is warts.

Only once in human history have we witnessed the total eradication of a dreaded disease, and that was smallpox more than two decades ago. Now humanity stands on the brink of a second: the global eradication of polio.

—From UNICEF's Polio Website

Chapters 16, 17, and 18 provide a review of the general biology of viruses and an introduction to basic virology. Chapter 38 continues this coverage by discussing viruses that are pathogenic to humans. Viruses are grouped according to their mode of acquisition and transmission, and viral diseases that occur in the United States are emphasized.

More than 400 different viruses can infect humans. Human diseases caused by viruses are unusually interesting, considering the small amount of genetic information introduced into a host cell. This apparent simplicity belies the severe pathological features, clinical consequences, and death that result from many viral diseases. With few exceptions, only prophylactic or supportive treatment is available. Collectively these diseases are some of the most common and yet most puzzling of all infectious diseases. The resulting frustration is compounded when year after year familiar diseases of unknown etiology or new diseases become linked to virus infections (table 38.1).

Table 38.1 Some Examples of Human Viral Diseases Recognized Since 1967

Year	Virus	Disease
1967	Marburg virus	Hemorrhagic fever
1973	Rotavirus	Major cause of infantile diarrhea worldwide
1975	Parvovirus	Aplastic crisis in chronic hemolytic anemia
1977	Ebola virus	Ebola hemorrhagic fever
1977	Hanta virus	Hemorrhagic fever with renal syndrome
1980	Human T-cell lymphotropic virus 1 (HTLV-1)	Adult T-cell leukemia
1982	Human T-cell lymphotropic virus 2 (HTLV-2)	Hairy-cell leukemia
1983	Human immunodeficiency virus (HIV)	Acquired immunodeficiency syndrome (AIDS)
1988	Human herpesvirus 6 (HHV-6)	Sixth disease (roseola subitum); may be associated with multiple sclerosis
1988	Hepatitis E	Enterically transmitted non-A, non-B hepatitis
1989	Hepatitis C	Parenterally transmitted non-A, non-B liver infection
1991	Guanarito virus	Venezuelan hemorrhagic fever
1992	Lymphocytic choriomeningitis virus	Central nervous system infection often leading to meningitis, encephalomyelitis, or other diseases
1993	Sin Nombre virus	Adult respiratory distress syndrome
1994	Sabia virus	Brazilian hemorrhagic fever
1994	Ross River virus	Ross River viral disease (Australia)
1995	Human herpesvirus 8 (HHV-8)	Associated with Kaposi's sarcoma in AIDS patients
1996	O'nyoung-nyong virus	Epidemic O'nyong fever
1997	Deer tick virus	Enzootic tick-borne encephalitis
1997	West Nile virus	West Nile fever
1997	Chicken flu (H5N1) virus	Influenza-like illness
1997	Transfusion-transmitted virus (TTV)	Hepatitis
1999	Australian bat lyssavirus (ABL)	ABL infection

38.1 Airborne Diseases

Because air does not support virus growth, any virus that is airborne must have originated from a source such as another human. When humans are the source of the airborne virus, it usually is propelled from the respiratory tract by an individual's coughing, sneezing, or vocalizing.

Chickenpox (Varicella) and Shingles (Zoster)

Chickenpox (varicella) is a highly contagious skin disease primarily of children 2 to 7 years of age. It is estimated that about 4 million cases occur annually in the United States. The causative agent is the varicella-zoster virus, a member of the family *Herpesviridae*, which is acquired by droplet inhalation into the respiratory system. The virus produces at least six glycoproteins that play a role in viral

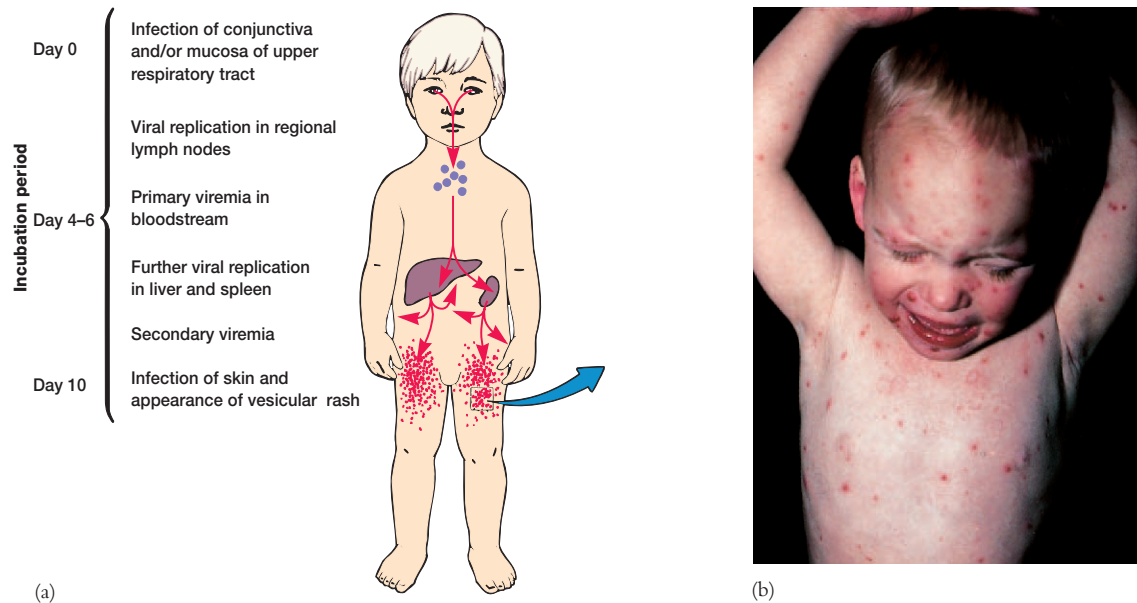


Figure 38.1 Chickenpox (Varicella). (a) Pathogenesis. (b) Typical vesicular skin rash. This rash occurs all over the body, but is heaviest on the trunk and diminishes in intensity toward the periphery.

attachment to specific receptors on respiratory epithelial cells, and their recognition by the human immune system results in humoral and cellular immunity. Following an incubation period of from 10 to 23 days, small vesicles erupt on the face or upper trunk, fill with pus, rupture, and become covered by scabs (**figure 38.1**). Healing of the vesicles occurs in about 10 days. During this time intense itching often occurs. Chickenpox can be prevented or the infection shortened with an attenuated varicella vaccine (Varivax; *see table 33.1*) or the drug acyclovir (Zovirax or Valtrex).

Individuals who recover from chickenpox are subsequently immune to this disease; however, they are not free of the virus, as viral DNA resides in a dormant (latent) state within the nuclei of cranial nerves and sensory neurons in the dorsal root ganglia. This viral DNA is maintained in infected cells but virions cannot be detected (**figure 38.2a**). When the infected person becomes immunocompromised by such factors as age, AIDS, or psychological or physiological stress, the viruses may become activated (*figure 38.2b*). They migrate down sensory nerves, initiate viral replication, and produce painful vesicles (*figure 38.2c*) because of sensory nerve damage. This syndrome is called **postherpetic neuralgia**. This reactivated form of chickenpox is termed **shingles (zoster)**. Most cases occur in people over 50 years of age. Shingles does not require specific therapy; however, in immunocompromised individuals, acyclovir (Zovirax or Valtrex; *see figure 35.14*), vidarabine (Vira-A), or famciclovir (Famvir) are recommended. More than 300,000 cases of zoster occur annually in the United States.

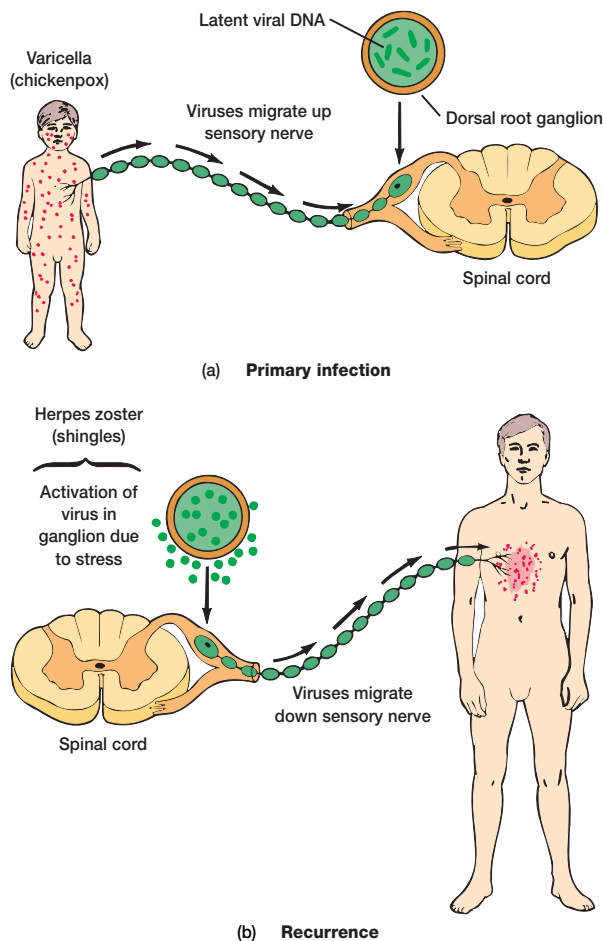
Influenza (Flu)

Influenza [Italian, to be influenced by the stars—*un influenza di freddo*], or the **flu**, is a respiratory system disease caused by orthomyxoviruses. Influenza viruses (*see figures 16.10i, 16.17a, b*) are

classified into A, B, and C groups based on the antigens (H and N) of their protein coats. One unique feature of the influenza viruses is the frequency with which changes in antigenicity occur. These changes are called antigenic variation. If the variation is small, it is called antigenic drift; if it is large, it is called antigenic shift. Antigenic variation occurs almost yearly with the influenza A virus, less frequently with the B virus, and has not been demonstrated with the C virus.

Animal reservoirs are critical to the epidemiology of human influenza. For example, rural China is one region of the world where chickens, pigs, and humans live in close, crowded conditions. Influenza is widespread in chickens; although chickens can't transmit the virus to humans, they can transfer it to pigs. Pigs can transfer it to humans and humans, back to pigs. Recombination between human and avian strains thus occurs in pigs, leading to major antigenic shifts. This explains why influenza continues to be a major epidemic disease and frequently produces worldwide pandemics. The worst pandemic on record occurred in 1918 and killed more than 20 million people. This disaster, traced to the Spanish influenza virus (*see figure 37.7*), was followed by pandemics of Asian flu (1957), Hong Kong flu (1968), and Russian flu (1977). (The names reflect popular impressions of where the episodes began, although all are now thought to have originated in China.)

The virus (*see figures 16.17a, 18.7, 18.9*) is acquired by inhalation or ingestion of virus-contaminated respiratory secretions. During an incubation period of 1 to 2 days, the virus adheres to the epithelium of the respiratory system (the neuraminidase present in envelope spikes may hydrolyze the mucus that covers the epithelium). The virus attaches to the epithelial cell by its hemagglutinin spike protein, causing part of the cell's plasma membrane to bulge inward, seal off, and form a vesicle (receptor-mediated endocytosis). This encloses the virus in an endosome. The hemagglutinin molecule in the virus envelope undergoes a dramatic conformational change when the endosomal pH decreases. The hydrophobic ends of the



(c)

Figure 38.2 Pathogenesis of the Varicella-Zoster Virus. (a) After an initial infection with varicella (chickenpox), the viruses migrate up sensory peripheral nerves to their ganglia, producing a latent infection. (b) When a person becomes immunocompromised or is under psychological or physiological stress, the viruses may be activated. (c) They migrate down sensory nerve axons, initiate viral replication, and produce painful vesicles. Since these vesicles usually appear around the trunk of the body, the name *zoster* (Greek for girdle) was originally used.

hemagglutinin spring outward and extend toward the endosomal membrane. After they contact the membrane, fusion occurs and the RNA nucleocapsid is released into the cytoplasmic matrix.

Influenza is characterized by chills, fever, headache, malaise, and general muscular aches and pains. These symptoms arise from the death of respiratory epithelial cells, probably due to attacks by activated T cells. Recovery usually occurs in 3 to 7 days, during which coldlike symptoms appear as the fever subsides. Influenza alone usually is not fatal. However, death may result from pneumonia caused by secondary bacterial invaders such as *Staphylococcus aureus*, *Streptococcus pneumoniae*, and *Haemophilus influenzae*. A commercially available identification technique is Directigen FLU-A (an enzyme immunoassay [EIA] rapid test). This test can detect influenza A virus in clinical specimens in less than 15 minutes.

As with many other viral diseases, only the symptoms of influenza usually are treated. However, the antiviral drugs amantadine (Symmetrel) (see figure 35.14), rimantadine (Flumadine), zanamivir (Relenza), and oseltamivir (Tamiflu) have been shown to reduce the duration and symptoms of type A influenza if administered during the first two days of illness. All four of these drugs attack the virus directly by plugging the catalytic site of the enzyme neuramidase. With the enzyme inactivated, viral particles can't travel from cell to cell. Aspirin (salicylic acid) should be

avoided in children younger than 14 years to reduce the risk of Reye's syndrome (**Box 38.1**). The mainstay for prevention of influenza since the late 1940s has been inactivated virus vaccines, (see table 33.1), especially for the chronically ill, individuals over age 65, residents of nursing homes, and health-care workers in close contact with people at risk. [Antiviral drugs \(pp. 821–22\).](#)

Currently three influenza A virus subtypes are epidemic in humans: those with H1N1, H2N1, and H3N2 hemagglutinin and neuraminidase surface glycoproteins. Influenza A infections usually peak in the winter and involve 10% or more of the population, with rates of 50 to 75% in school-age children. Influenza B usually accounts for only 3% of all flu cases in the United States.

Measles (Rubeola)

Measles [rubeola: Latin *rubeus*, red] is a highly contagious skin disease that is endemic throughout the world. The measles virus, a member of the genus *Morbillivirus* and the family *Paramyxoviridae*. The measles virus is monotypic, but small variations at the epitope level have been described. The variations are based on genetic variability in the virus genes. Such variations, however, have no effect on protective function since a measles infection still provides a lifelong immunity against reinfection. The virus

Box 38.1

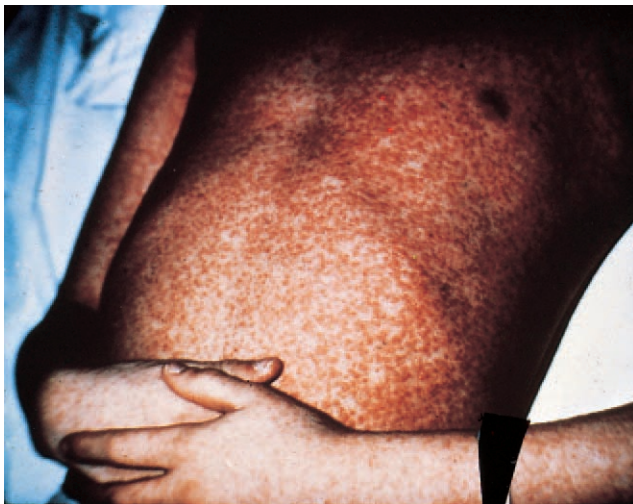
Reye's and Guillain-Barré Syndromes

An occasional complication of influenza, chickenpox, and a few other viral diseases in children under 14 years of age is **Reye's syndrome**. After the initial infection has disappeared, the child suddenly begins to vomit persistently and experiences convulsions followed by delirium and a coma. Pathologically, the brain swells with injury to the neuronal mitochondria, fatty infiltration into the liver occurs, blood ammonia is elevated, and both serum glutamic oxaloacetic transaminase (SGOT) and serum glutamic pyruvic transaminase (SGPT) are elevated in the blood. Diagnosis is made by the measurement of the levels of these enzymes and ammonia.

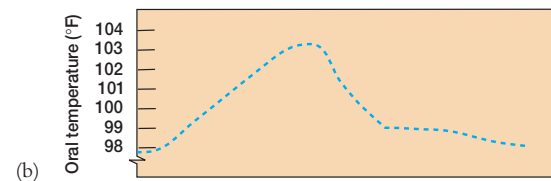
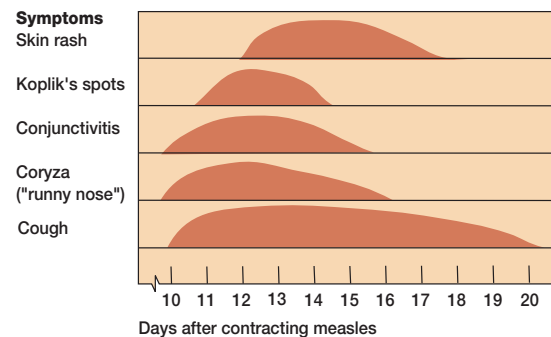
The relationship between the initial viral infection and the brain and liver damage is unknown. Treatment is nonspecific and directed toward reducing intracranial pressure and correcting metabolic and electrolyte abnormalities. Some children who recover have residual neurological

deficits—impaired mental capacity, seizures, and hemiplegia (paralysis on one side of the body). Mortality ranges from 10 to 40%. It is suspected that the use of aspirin or salicylate-containing products to lower the initial viral fever increases one's chances of acquiring Reye's syndrome.

Another condition that involves the central nervous system and is associated with influenza infections is **Guillain-Barré syndrome** (sometimes called **French polio**). In this disorder the individual suffers a delayed reaction (usually within 8 weeks) either to the actual virus infection or to vaccines against influenza. The virus or viral antigen in the vaccine damages the Schwann cells that myelinate the peripheral nerves and thus causes demyelination. As a result a prominent feature of this syndrome is a symmetric weakness of the extremities and sensory loss. Fortunately recovery usually is complete because the remaining undamaged Schwann cells eventually proliferate and wrap around the demyelinated nerves.



(a)



(b)

Figure 38.3 Measles (Rubeola). (a) The rash of small, raised spots is typical of measles. The rash usually begins on the face and moves downward to the trunk. (b) Signs and symptoms of a measles infection.

enters the body through the respiratory tract or the conjunctiva of the eyes. The receptor for the measles virus has recently been shown to be the complement regulator CD46, also known as membrane cofactor protein.

The incubation period is usually 10 to 21 days, and the first symptoms begin about the tenth day with a nasal discharge, cough, fever, headache, and conjunctivitis. Within 3 to 5 days skin eruptions occur as faintly pink maculopapular lesions that are at first discrete, but gradually become confluent (**figure 38.3**). The rash normally lasts about 5 to 10 days. Lesions of the oral cavity include the diagnostically useful bright-red **Koplik's spots** with a bluish-white

speck in the center of each. Koplik's spots represent a viral exanthem (a skin eruption) occurring in the form of macules or papules as a result of the viral infection. Very infrequently there occurs a progressive degeneration of the central nervous system called **subacute sclerosing panencephalitis** (table 38.6). No specific treatment is available for measles. The use of attenuated measles vaccine (Attenuvax) or in combination (MMR vaccine; measles, mumps, rubella) is recommended for all children (*see table 33.1*). Since public health immunization programs began in 1963, there has been over a 99% decrease in measles cases. Currently there are about 2,000 to 3,000 cases a year in the United States, and 90% of these



Figure 38.4 Mumps. Child with diffuse swelling of the salivary (parotid) glands due to the mumps virus.

are among unvaccinated individuals. In less well-developed countries, however, the morbidity and mortality in young children from measles infection remain high. It has been estimated that measles infects 50 million people and kills about 4 million a year worldwide. Serious outbreaks of measles are still reported in North America and Europe, especially among college students.

Mumps

Mumps is an acute generalized disease that occurs primarily in school-age children. The mumps virus is a member of the genus *Rubulavirus* in the family *Paramyxoviridae*. This virus (*see figure 16.10b*) is a pleomorphic, enveloped virus that contains a helical nucleocapsid composed of negative-sense, single-stranded RNA and three nucleocapsid-associated proteins (L, NP, and P). The virus is transmitted in saliva and respiratory droplets to nonimmune contacts. The portal of entry is the respiratory tract. The most prominent manifestations of mumps are swelling and tenderness of the salivary (parotid) glands 16 to 18 days after infection of the host by the virus (**figure 38.4**). The swelling usually lasts for 1 to 2 weeks and is accompanied by a low grade fever. Meningitis and inflammation of the epididymis and testes (**orchitis**) can be important complications associated with this disease—especially in the postpubescent male. Therapy of mumps is limited to symptomatic and supportive measures. A live, attenuated mumps virus vaccine is available. It usually is given as part of the triple MMR vaccine (*see table 33.1*). There are currently about 3,000 cases of mumps in the United States each year. Recently a few outbreaks have occurred among unvaccinated children in states without a comprehensive school immunization law. Prevention and control involves keeping children with mumps from school or other associated activities for about 2 weeks.

Respiratory Syndromes and Viral Pneumonia

Acute viral infections of the respiratory system are among the most common causes of human disease. The infectious agents are called the acute respiratory viruses and collectively produce a variety of clinical manifestations, including rhinitis (inflammation of the mucous membrane of the nose), tonsillitis, laryngitis, and bronchitis. The adenoviruses, coxsackievirus A, coxsackievirus B, echovirus, influenza viruses, parainfluenza viruses, poliovirus, respiratory syncytial virus, and reovirus are thought to be responsible. It should be emphasized that for most of these viruses there is a lack of specific correlation between the agent and the clinical manifestation—hence the term syndrome. Immunity is not complete, and reinfection is common. The best treatment is rest. [Disease syndromes, signs, and symptoms \(p. 850\); Classification of animal viruses \(pp. 399–402; appendix V\).](#)

In those cases of pneumonia for which no cause can be identified, viral pneumonia may be assumed if mycoplasmal pneumonia has been ruled out. The clinical picture is nonspecific. Symptoms may be mild, or there may be severe illness and death.

Respiratory syncytial virus (RSV) often is described as the most dangerous cause of lower respiratory infections in young children. In the United States over 90,000 infants are hospitalized each year and over 4,000 die. The RSV is a member of the RNA virus family, *Paramyxoviridae*. It is a negative single-stranded RNA virus that is enveloped with two virally specific glycoproteins as part of the structure. One of them, the large glycoprotein or G, is responsible for the binding of the virus to the host cell; the other, the fusion protein or F, permits fusion of the viral envelope with the host cell plasma membrane, leading to entry of the virus. The F protein also induces the fusion of the plasma membranes of infected cells. RSV thus gets its name from the resulting formation of a syncytium or multinucleated mass of fused cells. The multinucleated syncytia are responsible for inflammation, alveolar thickening, and the filling of alveolar spaces with fluid. The source of the RSV is hand contact and respiratory secretions of humans. Clinical manifestations consist of an acute onset of fever, cough, rhinitis, and nasal congestion. In infants and young children, this often progresses to severe bronchitis and viral pneumonia. Diagnosis is by either Directigen RSV or Test-Pack RSV rapid test kits. The virus is found worldwide and causes seasonal (November to May) outbreaks lasting several months. Treatment is with inhaled ribavirin (Virazole). A recently developed series of antibody (RSV-immune globulin) injections has been shown to reduce the severity of this disease in infants by 75%. Prevention and control consists of isolation for RSV-infected individuals, use of gowns when contact with secretions is likely, and strict attention to good handwashing practices.

Rubella (German Measles)

Rubella [Latin *rubellus*, reddish] was first described in Germany in the 1800s and was subsequently called **German measles**. It is a moderately contagious disease that occurs primarily in children 5 to 9 years of age. It is caused by the rubella virus, a single-stranded RNA virus that is a member of the family *Togaviridae*. Rubella is worldwide in distribution and occurs more frequently during the winter and spring months. This virus is spread in droplets that are



Figure 38.5 German Measles (Rubella). This disease is characterized by a rash of red spots. Notice that the spots are not raised above the surrounding skin as in measles (rubeola; see figure 38.3).

shed from the respiratory secretions of infected individuals. Once the virus is inside the body, the incubation period ranges from 12 to 23 days. A rash of small red spots (**figure 38.5**), usually lasting no more than 3 days, and a light fever are the normal symptoms. (This is why rubella is sometimes referred to as the “three-day measles.”) The rash appears as immunity develops and the virus disappears from the blood, suggesting that the rash is immunologically mediated and not caused by the virus infecting skin cells.

Rubella can be a disastrous disease (**congenital rubella syndrome**) in the first trimester of pregnancy and can lead to fetal death, premature delivery, or a wide array of congenital defects that affect the heart, eyes, and ears.

Because rubella is usually such a mild infection, no treatment is indicated. All children and women of childbearing age who have not been previously exposed to rubella should be vaccinated. The live attenuated rubella vaccine (part of MMR, *see table 33.1*) is recommended. Because routine vaccination began in the United States in 1969, fewer than 1,000 cases of rubella and 10 cases of congenital rubella occur annually.

Smallpox (Variola)

Smallpox (variola) was once one of the most prevalent of all diseases. The variola virus belongs to the family *Poxviridae*. The virion is large, brick-shaped, and contains a dumbbell-shaped core. It is of interest to note that the size of the smallpox virus is slightly larger than the size of the smallest bacteria—for example, *Chlamydia*. The genome inside the core consists of a single linear molecule of double-stranded DNA (*see figure 16.18*).

Variola was transmitted between humans by aerosol or contact. The virus produced a severe fever, prostration, and rash. Toxemia and septic shock led to case fatality rates of up to 30%. The infec-



Figure 38.6 Smallpox. Back of hand showing single crop of smallpox vesicles.

tion progressed from a skin rash that developed through stages of macule, papule, vesicle, and pustule (**figure 38.6**).

Since the advent of immunization with the vaccinia virus, and because of concerted efforts by the World Health Organization, the disease smallpox has been eradicated throughout the world. (The last case from a natural infection occurred in Somalia in 1977.) This was possible because a disease such as smallpox has obvious clinical features, virtually no asymptomatic carriers, only a human reservoir, and a short period of infectivity (3 to 4 weeks). Because humans are the only hosts, the spread of smallpox virus was prevented until no new cases developed, effectively eradicating the disease. The virus, however, is still kept in two locations: the CDC in Atlanta, Georgia, and the Russian State Center for Research on Virology and Biotechnology, Koltsovo, Novosibirsk region, Russia. [Disease \(smallpox\) and the early colonization of America \(p. 363\)](#)

1. Why are chickenpox and shingles discussed together? What is their relationship?
2. Briefly describe the course of an influenza infection and how the virus causes the symptoms associated with the flu. Why has it been difficult to develop a single flu vaccine?
3. What are some common symptoms of measles?
4. What are Koplik's spots?
5. What is one side effect that mumps can cause in a young postpubescent male?
6. Describe some clinical manifestations caused by the acute respiratory viruses.
7. Is viral pneumonia a specific disease? Explain.
8. When is a German measles infection most dangerous and why?

Box 38.2

Viral Hemorrhagic Fevers: A Microbial History Lesson

Scientists know of several viruses lurking in the tropics that—with a little help from nature—could wreak far more loss of life than will likely result from the AIDS pandemic. Collectively these viruses produce what is known as the **hemorrhagic fevers**. The viruses are passed among wild vertebrates, which serve as reservoir hosts. Arthropods transmit the viruses among vertebrates, and humans are infected when they invade the environment of the natural host. These diseases, distributed throughout the world, are known by over 70 names, usually denoting the geographic area where they were first described.

Viral hemorrhagic fevers can be fatal. Patients suffer headache, muscle pain, flushing of the skin, massive hemorrhaging either locally or throughout the body, circulatory shock, and death.

Despite the lack of public awareness, recent outbreaks may foreshadow much broader outbreaks in the future. For example, in the late 1960s dozens of scientists in West Germany fell seriously ill, and several died, from a mysterious new disease. Victims suffered from a breakdown of liver function and a bizarre combination of bleeding and blood clots. The World Health Organization traced the outbreak to a batch of fresh monkey cells the scientists had used to grow polioviruses. The cells from the imported Ugandan monkeys were infected with the lethal tropical Marburg virus and the scientists suffered from **Marburg viral hemorrhagic fever**.

In 1977 the *Plebovirus* causing Rift Valley Fever in sheep and cattle moved from these animals into the South African population. The virus, which causes severe weakness, incapacitating headaches, damage to the retina, and hemorrhaging, then made its way to Egypt, where millions of humans became infected and thousands died.

Among the most frightening hemorrhagic outbreak was that of the **Ebola virus hemorrhagic fever** in Zaire and Sudan in 1976. This disease infected more than 1,000 people and left over 500 dead. It became concentrated in hospitals, where it killed many of the Belgian physicians

and nurses treating infected patients. A similar outbreak occurred in this same area in 1995 but was quickly contained.

In the United States in 1989, epidemiologists provided new evidence that rats infected with a potentially deadly hemorrhagic virus are prevalent in Baltimore slums. The virus appears to be taking a previously unrecognized toll on the urban poor by causing **Korean hemorrhagic fever**.

In the summer of 1993, reports appeared in the news media about a mysterious illness that had caused over 30 deaths among the Navajo nation in the four-corners area of the southwestern United States. The CDC finally determined the causative agent to be a hantavirus, a negative single-stranded RNA virus that is a member of the family *Bunyaviridae*. Hantaviruses are endemic in rodents, such as deer mice, in many areas of the world. Deer mice shed the virus in their saliva, feces, and urine. Humans contract the disease when they inhale aerosolized particles containing the excreted virus. Throughout Asia and central Europe, hantaviruses cause hemorrhagic fever with renal syndrome in humans. But the type of virus found in the Southwest had not been previously recognized, and no hantavirus anywhere in the world has been associated with the clinical syndrome initially seen among the Navajo; namely, the **hantavirus pulmonary syndrome** in which the virus destroys the lungs. In 1993 the CDC named this virus **pulmonary syndrome hantavirus** (sometimes called the Sin Nombre or no-name virus) and to date, isolated cases have been reported from almost every state. Prevention involves wearing gloves when handling mice and spraying the feces and urine of all mice with a disinfectant.

Although to date, these epidemics have not become global, they do provide a humbling vision of humankind's viral vulnerability. History shows that the life-threatening viral hemorrhagic outbreaks often have arisen when humans moved into unexplored terrain or when living conditions deteriorated in ways that generated new viral hosts. In each case medical and scientific resources have been reactive, not proactive.

Table 38.2 Summary of the Seven Major Human Arbovirus Diseases That Occur in the United States

Disease	Distribution	Vectors	Mortality Rate
California encephalitis (La Crosse)	North Central, Atlantic, South	Mosquitoes (<i>Aedes</i> spp.)	Fatalities rare
Colorado tick fever	Pacific Coast (mountains)	Ticks (<i>Dermacentor andersoni</i>)	Fatalities rare
Eastern equine encephalitis (EEE)	Atlantic, Southern Coast	Mosquitoes (<i>Aedes</i> spp.)	50–70%
St. Louis encephalitis (SLE)	Widespread	Mosquitoes (<i>Culex</i> spp.)	10–30%
Venezuelan equine encephalitis (VEE)	Southern United States	Mosquitoes (<i>Aedes</i> spp. and <i>Culex</i> spp.)	20–30% (children) <10% (adults)
Western equine encephalitis (WEE)	Mountains of the West	Mosquitoes (<i>Culex</i> spp.)	3–7%
West Nile encephalitis	Most of Atlantic Coast	Mosquitoes (various spp.)	?

38.2 Arthropod-Borne Diseases

The arthropod-borne viruses (arboviruses) are transmitted by bloodsucking arthropods from one vertebrate host to another. They multiply in the tissues of the arthropod without producing disease, and the vector acquires a lifelong infection. Approximately 150 of the recognized arboviruses cause illness in hu-

mans. Diseases produced by the arboviruses can be divided into three clinical syndromes: (1) fevers of an undifferentiated type with or without a rash; (2) encephalitis (inflammation of the brain), often with a high fatality rate; and (3) hemorrhagic fevers, also frequently severe and fatal (**Box 38.2**). **Table 38.2** summarizes the seven major human arbovirus diseases that occur in the United States. For all these diseases, immunity is believed to be

permanent after a single infection. No vaccines are available for the human arthropod-borne diseases listed in table 38.2, although supportive treatment is beneficial.

Colorado Tick Fever

Colorado tick fever is caused by an RNA virus of the genus *Coltivirus*. The tick, *Dermacentor andersoni*, is the main vector that transmits the disease to humans. Ground squirrels, rabbits, and deer are the principal reservoirs. The disease occurs in mountainous regions in western states. Approximately 300 cases are reported annually in the United States, but the actual incidence is probably much higher. After a 3 to 6 day incubation period, there is an abrupt onset of fever, chills, severe headache, photophobia, rash, and muscle pain. These symptoms persist for 5 to 8 days and then disappear; complications are rare. Serology is used to confirm the diagnosis. No specific treatments exist for Colorado tick fever; therapy is limited to supportive care. Prevention involves common tick precautions (see pp. 910–11).

Yellow Fever

Yellow fever is not now endemic to the United States, but because of its historical importance, it is briefly discussed here. Yellow fever was the first human disease found to be caused by a virus. (Walter Reed discovered this in 1901.) It also provided the first confirmation (by Carlos Juan Finley) that an insect could transmit a virus. Yellow fever is caused by a flavivirus that is endemic in many tropical areas, such as Mexico, South America, and Africa.

The disease received its first name, yellow jack, because jaundice is a prominent sign in severe cases. The jaundice is due to the deposition of bile pigments in the skin and mucous membranes because of damage to the liver. The disease is spread through a population in two epidemiological patterns. In the urban cycle human-to-human transmission is by *Aedes aegypti* mosquitoes. In the sylvatic cycle the mosquitoes transmit the virus between monkeys and from monkeys to humans (sylvatic means in the woods or affecting wild animals).

Once inside a person the virus spreads to local lymph nodes and multiplies; from this site it moves to the liver, spleen, kidneys, and heart, where it can persist for days. In the early stages of the disease, the infected person experiences fever, chills, headache, and backache, followed by nausea and vomiting. In severe cases the virus produces lesions in the infected organs and hemorrhaging occurs.

There is no specific treatment for yellow fever. Diagnosis is by serology. An active immunity to yellow fever results from an initial infection or from vaccines containing the attenuated yellow fever 17D strain or the Dakar strain virus. Prevention and control of this disease involves vaccination (see table 33.1) and control of the insect vector.

38.3 Direct Contact Diseases

Acquired Immune Deficiency Syndrome (AIDS)

It is now recognized that **AIDS (acquired immune deficiency syndrome)** is the first great pandemic of the second half of the twentieth century. First described in 1981 AIDS is the result of an

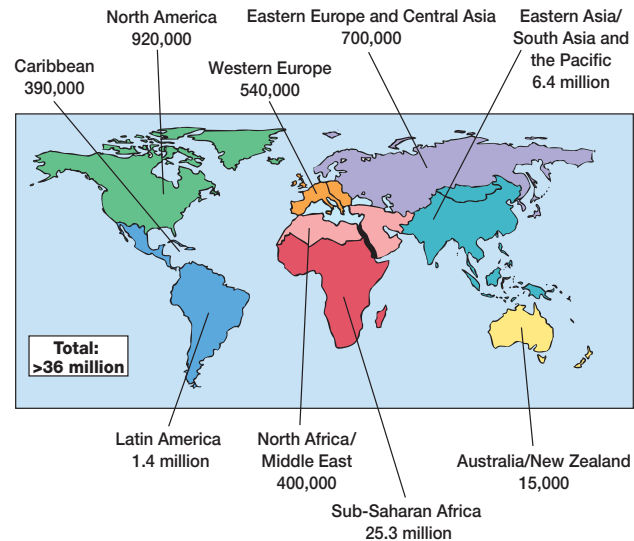


Figure 38.7 Distribution of HIV/AIDS in Adults by Continent or Region. The figure shows data from a 2000 United Nations report. According to recent estimates by the UN, the number of HIV/AIDS cases may be over 36 million. Source of data: UNAIDS.

infection by the **human immunodeficiency virus (HIV)**, a lentivirus within the family *Retroviridae*. The disease appears to have begun in central Africa as early as the 1950s; HIV may have developed in the human population in the 1930s, or even earlier. Simian immunodeficiency viruses (SIVs) related to HIV-1 and HIV-2, the strains primarily responsible for AIDS, have been isolated from African primates. The SIV from chimpanzees seems to have infected humans and developed into HIV-1; HIV-2 may have arisen from the SIV that infects sooty mangabeys. Once established, HIV-1 spread to the Caribbean and then to the United States and Europe.

Epidemiologically AIDS occurs worldwide (figure 38.7). The groups most at risk in acquiring AIDS are (in descending order of risk) homosexual/bisexual men; intravenous (IV) drug users; heterosexuals who have intercourse with drug users, prostitutes (sex trade workers), and bisexuals; transfusion patients or hemophiliacs who must receive clotting factor preparations made from donated blood; and children born of infected mothers. The mortality rate from AIDS is extremely high.

In the United States, AIDS is caused primarily by the HIV-1 virus (some cases result from an HIV-2 infection). This virus is a retrovirus and closely related to HTLV-1, the cause of adult T-cell leukemia, and HTLV-2, which has been isolated from individuals with hairy-cell leukemia (see leukemia, pp. 887–88). HIV-1 is an enveloped lentivirus and a member of the family *Retroviridae* (see section 18.1) with a cylindrical core inside its capsid (figure 38.8). The core contains two copies of its plus single-stranded RNA genome and several enzymes. Thus far 10 virus-specific proteins have been discovered. One of them, the gp120 envelope protein, participates in HIV-1 attachment to CD4⁺ cells (T-helper cells; see figure 38.9).

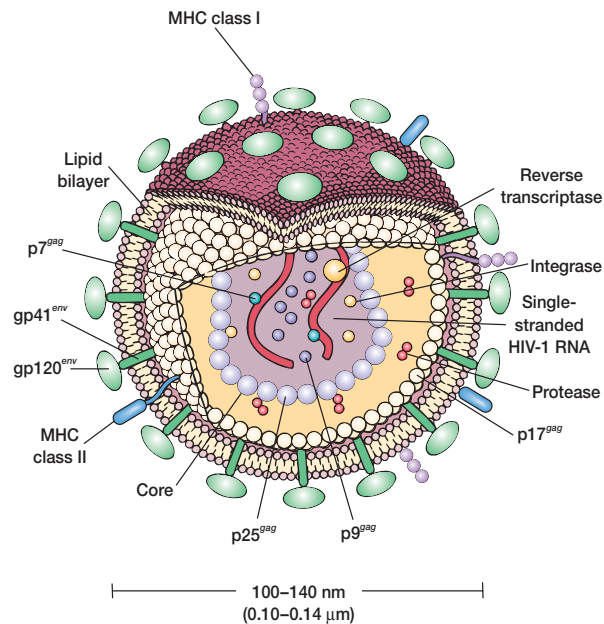


Figure 38.8 Schematic Diagram of the HIV-1 Virion. The HIV-1 virion is an enveloped structure containing 72 external spikes. These spikes are formed by the two major viral-envelope proteins, gp120 and gp41. (Gp stands for glycoprotein—the proteins are linked to sugars—and the number refers to the mass of the protein, in thousands of daltons.) The HIV-1 lipid bilayer is also studded with various host proteins, including class I and class II major histocompatibility complex molecules, acquired during virion budding. The cone-shaped core of HIV-1 contains four nucleocapsid proteins (p25, p17, p9, p7) each of which is proteolytically cleaved from a 53 kDa *gag* precursor by the HIV-1 protease. The phosphorylated p25 polypeptide forms the chief component of the inner shell of the nucleocapsid, whereas the p17 protein is associated with the inner surface of the lipid bilayer and stabilizes the exterior and interior components of the virion. The p7 protein binds directly to the genomic RNA through a zinc-finger structural motif and together with p9 forms the nucleoid core. The retroviral core contains two copies of the single-stranded HIV-1 genomic RNA that is associated with the various preformed viral enzymes, including the reverse transcriptase, integrase, ribonuclease, and protease.

The AIDS virus is acquired by direct exposure of a person's bloodstream to body fluids (blood, semen, vaginal secretions) containing the virus, through sexual contact, or perinatally from an infected mother to her fetus. It also is possible that a newborn can be infected through breast-feeding. Once inside the body, the virus gp120 envelope protein (figure 38.8) binds to the CD4 glycoprotein plasma membrane receptor on CD4⁺ T cells, macrophages, dendritic cells, and monocytes (figure 38.9). (Dendritic cells are present throughout the body's mucosal surfaces and bear the CD4 protein. Thus it is possible that these are the first cells infected by HIV in sexual transmission.) Recent evidence shows that the virus requires a coreceptor in addition to the CD4 receptor. Macrophage-tropic strains, which seem to predominate early in the disease and infect both macrophages

and T cells, require the CCR5 (CC-CKR-5) chemokine receptor protein as well as CD4. A second chemokine coreceptor, called CXCR-4 or fusin, is T cell-tropic and used by an HIV strain that is active at later stages of the infection. This strain induces the formation of syncytia, as described later. Individuals with two defective copies of the *CCR5* gene do not seem to get AIDS; apparently the virus cannot infect their T cells. People with one good copy of the *CCR5* gene do get AIDS but survive several years longer than those with no mutation. [Host cell receptors and virion adsorption \(pp. 399–403\); Replication and transcription in retroviruses \(p. 407\).](#)

Entry into the host cell begins when the envelope fuses with the plasma membrane, and the virus releases its core and two RNA strands into the cytoplasm. Inside the infected cell, the core protein remains associated with the RNA as it is copied into a single strand of DNA by the RNA/DNA-dependent DNA polymerase activity of the reverse transcriptase enzyme. The RNA is next degraded by another reverse transcriptase component, ribonuclease H, and the DNA strand is duplicated to form a double-stranded DNA copy of the original RNA genome. A complex of the double-stranded DNA (the provirus) and the integrase enzyme moves into the nucleus. Then the proviral DNA is integrated into the cell's DNA through a complex sequence of reactions catalyzed by the integrase (figure 38.9). The integrated provirus can remain latent, giving no sign of its presence. Alternatively the provirus can force the cell to synthesize viral mRNA. Some of the RNA is translated to produce viral proteins by the cell's own ribosomes. Viral proteins and the complete HIV-1 RNA genome are then assembled into new virions that bud from the infected host cell (see figure 18.10). Eventually the host cell lyses.

The precise mechanism of AIDS pathogenesis still is not known, and many hypotheses exist. Many believe that AIDS is caused primarily by depletion of T cells and disruption of their function, although the exact mechanisms are unclear. The cytopathic effect may be due to the disruption of plasma membrane permeability and function by excessive virus budding. Free gp120 proteins may bind to CD4 proteins on uninfected cells, making them targets for attack by immune system cells. Infected cells do fuse with other cells to form large, multinucleate syncytia that eventually die, and this may contribute greatly to cell destruction. Possibly, insertion of the provirus DNA into the cell's genome and the transposition of the integrated provirus disrupt cell function and destroy the host T cell. Recent research shows that the virus replicates at a furious pace, about a billion virus particles a day. The immune system often manages to hold it off for years but eventually succumbs in almost all cases.

Once a human's CD4⁺ cells are infected with HIV-1, four types of pathological changes may ensue. First, a mild form of AIDS may develop with symptoms that include fever, malaise, headache, macular rash, weight loss, lymph node enlargement (lymphadenopathy), oral candidiasis (figure 38.10a), and the presence of antibodies to HIV-1 (figure 38.11). These symptoms may occur in the first few months after infection, last for 1 to 3 weeks, and recur. This is known as **AIDS-related complex (ARC)**. ARC develops into a full-blown case of AIDS in an undetermined proportion of cases.

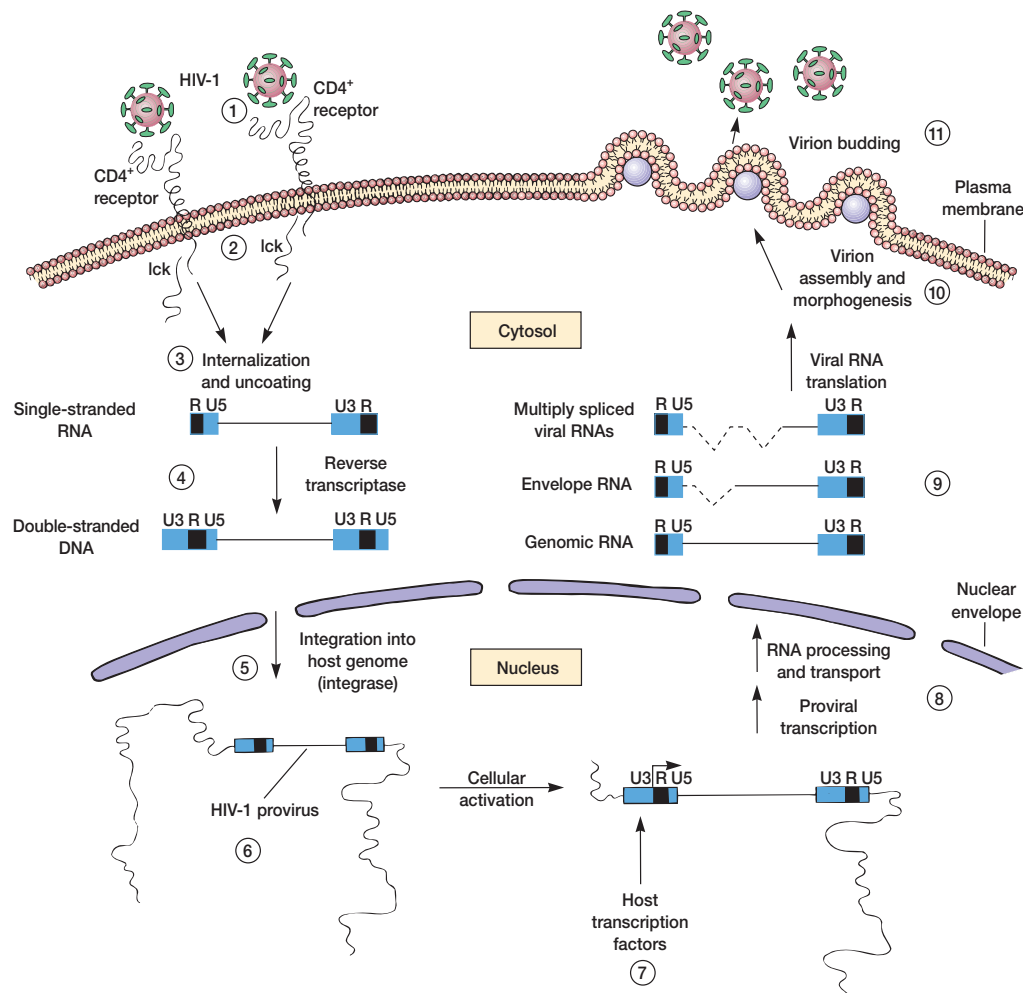


Figure 38.9 Life Cycle of HIV-1. (1) After interaction of gp120 with the CD4⁺ cell plasma membrane receptor, gp41-mediated membrane fusion occurs. (2) This leads to the entry of HIV-1 into the cell. The Lck denotes a lymphoid-specific tyrosine kinase that binds to CD4. (3) After internalization and uncoating, reverse transcription of viral RNA begins. (4) The double-stranded DNA form of the virus genome is produced in the presence of appropriate host factors. (5) The HIV-1 integrase promotes the insertion of this viral DNA duplex into the CD4 cell's genome after the DNA has entered the nucleus. (6) This gives rise to the HIV-1 provirus. (7) The expression of the HIV-1 gene is stimulated initially by the action of specific inducible and constitutive host transcription factors with binding sites in the long terminal repeat. Their binding leads to the sequential production of various viral mRNAs. (8) The first mRNAs produced correspond to the multiply spliced species of approximately 2.0 kilobases encoding *tat*, *rev*, and *nef* regulatory proteins. (9) Subsequently the viral structural proteins are produced, allowing the (10) assembly and morphogenesis of the virions. (11) The new HIV-1 virions that are produced by viral budding from the host CD4⁺ cell can then reinitiate the retroviral life cycle by infecting other CD4⁺ target cells.



(a)



(b)

Figure 38.10 Some Diseases Associated with AIDS. (a) Candidiasis of the oral cavity and tongue (thrush) caused by *Candida albicans*. (b) Kaposi's sarcoma on the arm of an AIDS patient. The flat purple tumors can occur in almost any tissue and are frequently multiple.

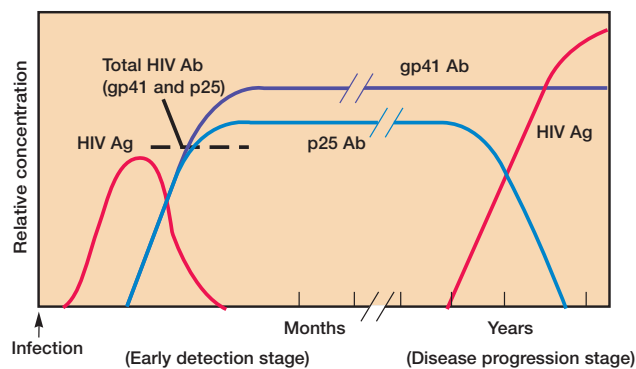


Figure 38.11 The Typical Serological Pattern in an HIV-1 Infection. HIV-1 antigen (HIV Ag) is detectable as early as 2 weeks after infection and typically declines at seroconversion. Seroconversion occurs when HIV-1 antibodies have risen to detectable levels. This usually takes place several weeks to months after the HIV-1 infection. The period between HIV-1 infection and seroconversion often is associated with an acute illness. Whether or not the individual has flu-like symptoms, the appearance of circulating HIV-1 antigens typically occurs before IgG antibodies against gp41 and gp25 develop. HIV-1 antigen then usually disappears following seroconversion but reappears in the latter stages of the disease. The reappearance of antigen usually indicates impending clinical deterioration. An asymptomatic HIV-1 antigen-positive individual is six times more likely to develop AIDS within 3 years than a similar individual who is HIV-1 antigen negative. Thus testing for the presence of the HIV-1 antigen assists clinicians in monitoring the progression of the disease.

Second, a true case of AIDS can develop directly upon infection. The mean interval between HIV infection and the onset of AIDS appears to be about 8 to 10 years, although it varies considerably with each individual. At first, a person's immune system responds to the HIV-1 infection by manufacturing HIV-1 antibodies,

but not in sufficient quantities to stop the viral attack. The virus becomes established within primarily CD4⁺ T-helper cells, and HIV accumulates in lymphoid organs in large quantities even before symptoms appear. Initially, CD4⁺ T-helper cells proliferate abnormally in the lymph nodes. Thereafter the lymph nodes' internal structure collapses due to viral replication. This leads to a decline in the number of lymphocytes within the lymph nodes and results in a selective depletion of the CD4⁺ T-cell subset that is critical to the propagation of the entire T-cell pool. When this CD4⁺ population declines, interleukin-2 (IL-2) production also decreases. Because IL-2 stimulates the production of T cells in general, the whole T-cell population may decline. This leaves the infected person open to opportunistic infections: invasion by pathogens that proliferate widely only because the immune system is defective.

It should be noted that factors other than direct T-cell destruction also may be involved in AIDS pathogenesis. HIV may reduce the immune response by destroying or disabling dendritic cells, which present foreign antigens to T cells. HIV also mutates exceptionally rapidly and thus could evade and eventually overwhelm the immune system. HIV may disrupt the balance between different types of T-helper cells and consequently decrease the killer T-cell population. It is possible that several different mechanisms contribute to T-cell destruction.

New findings suggest still another potential mechanism for the depletion of CD4⁺ cells (**figure 38.12**). In HIV-infected individuals the loss of CD4⁺ cells is associated with lymphocyte activation; however, this activation does not result in cell proliferation, as it does normally, but rather in cell death by a mechanism known as programmed cell death [**apoptosis**; Greek "a falling off" from *apo*, off, and *ptosis*, fall].

In 1993 the CDC revised its definition of AIDS to include all HIV-infected persons who have fewer than 200 CD4⁺ cells per μ l or a CD4⁺ cell percentage of total lymphocytes of less than 14. The reason for this definition is that the development of a particular opportunistic infection is related to the concentration of CD4⁺ cells in

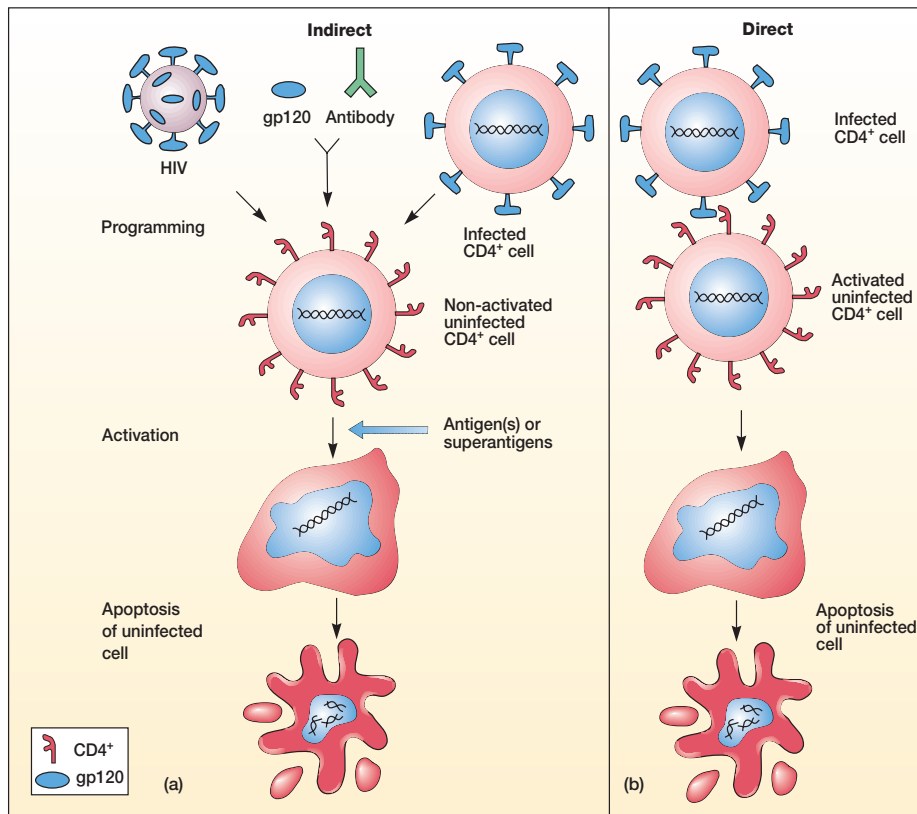


Figure 38.12 Apoptosis and AIDS. Apoptosis is a homeostatic physiological suicide mechanism in which cell death occurs naturally during normal tissue turnover. Usually apoptosis occurs after activation of a calcium-dependent endogenous endonuclease. Cells undergoing apoptosis display profound structural changes such as a decrease in cell volume, blebbing of the plasma membrane, and nuclear fragmentation. The nuclear DNA is cleaved into short oligonucleosomal length DNA fragments. The dying cell sheds small membrane-bound apoptotic bodies, which are phagocytosed and digested. **(a)** There may be several ways in which an HIV infection can indirectly trigger apoptosis. In all cases an initial event would program or prime the target cell so that apoptosis would be triggered by the binding of antigens or superantigens to the cell's T-cell receptors. Possibly the external gp120 envelope glycoprotein of the HIV virion binds to the CD4 protein on lymphocytes and programs the lymphocyte. A combination of free gp120 and antibodies to gp120 also could stimulate programmed cell death. First, the gp120 would bind to CD4 receptors. Then antibodies would attach to the gp120 and cause clustering of the receptors, thus priming the uninfected CD4⁺ cell. It also is possible that binding of the infected cell's surface gp120 proteins to the CD4 receptors on an uninfected cell will program the uninfected cell for apoptosis in response to antigens. **(b)** Apoptosis may be directly triggered in an uninfected cell. The gp120 envelope proteins on the surface of an infected cell may combine with the CD4 proteins of an uninfected cell and directly stimulate programmed cell death without activation by antigens.

the blood. Healthy individuals have about 1,000 such cells in every μl of blood. In HIV-1 infected individuals, the number declines by an average of about 40 to 80 cells per μl each year. When the CD4⁺ cell count falls to between 400 and 200 per μl , the first opportunistic infections and disease processes usually appear (**table 38.3**). Examples of such opportunistic infections and diseases include *Pneumocystis carinii* pneumonia; *Mycobacterium avium-intracellulare* pneumonia (see pp. 902–3); toxoplasmosis; herpes zoster infection (**figure 38.2b**); chronic diarrhea caused by *Cyclospora*; cryptococcal meningitis; *Histoplasma capsulatum* infection; and tuberculosis (see **figure 39.7**).

The third main type of disease caused by HIV-1 involves the central nervous system since virus-infected macrophages can cross the blood-brain barrier. The classical symptoms of central nervous system disease in AIDS patients are headaches, fevers, subtle cognitive changes, abnormal reflexes, and ataxia (irregularity of muscular action). Dementia and severe sensory and motor changes characterize more advanced stages of the disease. Autoimmune neuropathies, cerebrovascular disease, and brain tumors also are common. Histological changes include inflammation of neurons, nodule formation, and demyelination. All evidence indicates that these neurological changes are correlated with higher levels of HIV-1

Table 38.3 Disease Processes Associated with AIDS

Candidiasis of bronchi, trachea, or lungs
Candidiasis, esophageal
Cervical cancer, invasive
Coccidioidomycosis, disseminated or extrapulmonary
Cryptosporidiosis, chronic intestinal (>1 month's duration)
<i>Cyclospora</i> , diarrheal disease
Cytomegalovirus disease (other than liver, spleen, or lymph nodes)
Cytomegalovirus retinitis (with loss of vision)
Encephalopathy, HIV-related
Herpes simplex: chronic ulcer(s) (>1 month's duration); or bronchitis, pneumonitis, or esophagitis
Histoplasmosis, disseminated or extrapulmonary
Isosporiasis, chronic intestinal (>1 month's duration)
Kaposi's sarcoma
Lymphoma, Burkitt's (or equivalent term)
Lymphoma, immunoblastic (or equivalent term)
Lymphoma, primary, of brain
<i>Mycobacterium avium</i> complex or <i>M. kansasii</i>
<i>Mycobacterium tuberculosis</i> , any site
<i>Mycobacterium</i> , other species or unidentified species
<i>Pneumocystis carinii</i> pneumonia
Pneumonia, recurrent
Progressive multifocal leukoencephalopathy
<i>Salmonella</i> septicemia, recurrent
Toxoplasmosis of brain
Wasting syndrome due to AIDS

Source: Data from *MMWR* 41 (No. RR17). 1993 Revised Classification System for HIV Infection and Expanded Surveillance Case Definition for AIDS Among Adolescents and Adults.

antigen (figure 38.11) and/or the HIV-1 genome in central nervous system tissue. In AIDS dementia, macrophages and glial cells (supporting cells of the nervous system) are primarily infected and bud new viruses. However, it is unlikely that direct infection of these cells by HIV-1 is responsible for the symptoms. More probably the symptoms arise through either the secretion of viral proteins or viral induction of cytokines that bind to glial cells and neurons. HIV-1 induction of interleukin-1 and tumor necrosis factor- α (TNF- α) may stimulate further viral reproduction and the induction of other cytokines (e.g., interleukin-6, granulocyte-macrophage colony-stimulating factor [GM-CSF]). IL-1 and TNF- α in combination with IL-6 and GM-CSF could account for the many clinical and histopathological findings in the central nervous system of AIDS individuals.

The fourth result of an HIV infection is cancer. Individuals infected with HIV-1 have an increased risk of three types of tumors: (1) Kaposi's sarcoma (figure 38.10b), (2) carcinomas of the mouth and rectum, and (3) B-cell lymphomas or lymphoproliferative disorders. It seems likely that the depression of the initial immune response enables secondary tumor-causing agents to initiate the cancers.

In 1994 it was discovered that Kaposi's sarcoma-associated herpesvirus (KSHV, also known as human herpesvirus 8, or HHV-8) is a virus that is consistently present in Kaposi's sarcoma and in primary effusion (body cavity-based) lymphomas. These cancers occur most frequently in AIDS patients. KSHV is a

gamma herpesvirus with homology to herpesvirus saimiri and Epstein-Barr virus, both of which can transform lymphocytes.

The laboratory diagnosis of AIDS can be by viral isolation and culture or by using assays for viral reverse transcriptase activity or viral antigens (figure 38.11). However, diagnosis is best accomplished through the detection of specific anti-HIV antibodies in the blood. These antibodies may be detected by GENIE or HIVAG-1 (Abbott) rapid tests, the SUDS (Murex) 10 minute assay, enzyme-linked immunoabsorbent assay, indirect immunofluorescence, immunoblot (Western Blot), and radioimmunoprecipitation methods. The most sensitive HIV assay employs the polymerase chain reaction (*see pp.* 326–27). PCR can be used to amplify and detect tiny amounts of viral RNA and cDNA in virions or infected host cells. With proper procedures quantitative PCR assays will provide an estimate of a patient's viral load. This is particularly significant because the level of virions in the blood, as well as the concentration of CD4⁺ cells, is very predictive of the clinical course of the infection. The probable time to development of AIDS can be estimated from the patient's blood virion level and CD4⁺ cell count.

At present there is no complete cure for AIDS. Primary treatment is directed at reducing the viral load and disease symptoms, and at treating opportunistic infections and malignancies. The antivirals currently approved for use in HIV disease are of three types. (1) Most reverse transcriptase inhibitors are nucleoside analogues that inhibit the enzyme reverse transcriptase as it synthesizes DNA. Examples include AZT or zidovudine (Retrovir), didanosine (Videx), ddC or zalcitabine (HIVID), stavudine (Zerit), and lamivudine or 3TC (Epivir). (2) The nonnucleoside inhibitors of reverse transcriptase include delavirdine (Rescriptor) and nevirapine (Viramune). (3) The protease inhibitors work by blocking the activity of the HIV protease and thus interfere with virion assembly. Examples include indinavir (Crixivan), zalcitabine (Norvir), nelfinavir (Viracept), and saquinavir (Invirase). The most successful treatment approach is to use drug combinations. An effective combination is a cocktail of AZT, lamivudine, and a protease inhibitor such as ritonavir. In many patients the virus disappears from the patient's blood with proper treatment and drug-resistant strains do not seem to arise. Recently it has been discovered that HIV is dormant in the memory T cells, survives the drug cocktail, and is capable of reactivation. Thus patients cannot be completely cured with this drug treatment protocol. It should be noted that side effects can be very severe, and not everyone can tolerate the drugs. If the treatment is not started soon enough, the immune system does not fully recover after HIV levels have fallen.

Another avenue of current research is the development of a vaccine that can (1) stimulate the production of neutralizing antibodies which can bind to the virus envelope and prevent it from entering host cells, and (2) promote the destruction of those cells already infected with the virus. The production of an effective vaccine, if possible, is not yet in sight. One difficulty is that the envelope proteins of the virus (figure 38.8) continually change their antigenic properties.

Prevention and control of AIDS involves screening of blood and heat treatment of blood products to destroy the virus. Education and protected sexual behavior and practices, including the use of



Figure 38.13 Cold Sores. Herpes simplex fever blisters on the lip, caused by herpes simplex type 1 virus.

condoms, are keys to comprehensive community-based prevention programs. Education of intravenous drug users concerning the need to avoid sharing needles and syringes is also very important in prevention programs. It must be emphasized that protected sexual behavior and the use of condoms is safer but not totally safe. The only absolute protection against AIDS and other sexually transmitted diseases is abstinence and completely monogamous relationships.

Cold Sores

Cold sores or fever blisters (herpes labialis) are caused by the herpes simplex virus type 1 (HSV-1). Like all herpesviruses, it is a double-stranded DNA virus with an enveloped, icosahedral capsid. The term herpes is derived from the Greek word meaning “to creep,” and clinical descriptions of herpes labialis (lips) go back to the time of Hippocrates (circa 400 B.C.). Transmission is through direct contact of epithelial tissue surfaces with the virus (*see figure 18.5*). A blister(s) develops at the inoculation site (**figure 38.13**) because of host- and viral-mediated tissue destruction. Most blisters involve the epidermis and surface mucous membranes of the lips, mouth, and gums (**gingivostomatitis**). The blisters generally heal within a week. However, after a primary infection, the virus travels to the trigeminal nerve ganglion, where it remains in a latent state for the lifetime of the infected person. Stressful stimuli such as excessive sunlight, fever, trauma, chilling, emotional stress, and hormonal changes can reactivate the virus. Once reactivated, the virus moves from the trigeminal ganglion down a peripheral nerve to the border of the lip or other parts of the face to produce another fever blister. Primary and recurring infections also may occur in the eyes, causing **herpetic keratitis** (inflammation of the cornea)—currently a major cause of blindness in the United States. The drugs vidarabine (Vira-A) and acyclovir (Zovirax) are effective against cold sores. By adulthood, 70 to 90% of all people in the United States have been infected and have type 1 herpes antibodies. The “gold standard” for the detection of herpes is cell culture. However, cell culture requires up to 14 days for results. The rapid detection of herpes simplex virus

(HSV) antigen is by several enzyme immunoassay kits (HER-PCHEK, VIDAS, IDEIA, SureCell). This rapid test is especially useful in cases of pregnant women with genital infections and individuals who are particularly susceptible to severe infections.

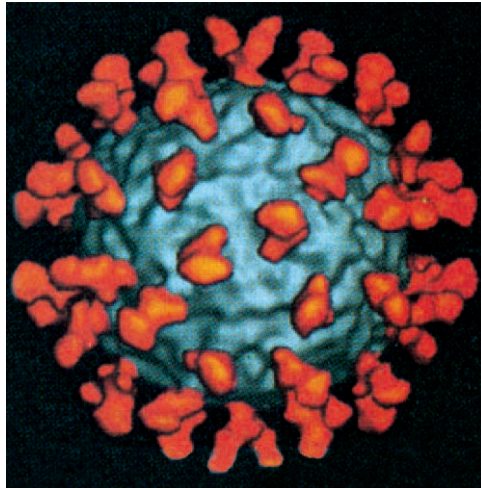
Common Cold

The **common cold** [**coryza**: Greek *koryza*, discharge from the nostrils] is one of the most frequent infections experienced by humans of all ages. About 50% of the cases are caused by rhinoviruses [Greek *rhinos*, nose], single-stranded RNA viruses in the family *Picornaviridae* (*see figure 16.10k*). There are over 115 distinct serotypes, and each of these antigenic types has a varying capacity to infect the nasal mucosa and to cause a cold. In addition, immunity to many of them is transitory. Several other respiratory viruses are also associated with colds (e.g., coronaviruses and parainfluenza viruses). Thus colds are common because of the diversity of rhinoviruses, the involvement of other respiratory viruses, and the lack of a durable immunity.

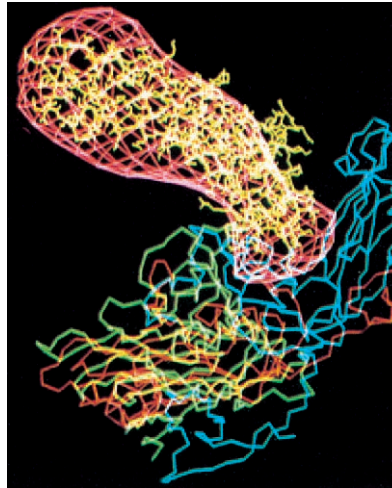
Rhinoviruses provide an excellent example of the medical relevance of research on virus morphology. The complete rhinovirus capsid structure has recently been elucidated with the use of X-ray diffraction techniques. The results help explain rhinovirus resistance to human immune defenses. Human antibodies against the virus combine with four types of surface proteins. Unfortunately there are at least 89 variants of these four antigens, making the production of a cold vaccine from them unlikely. The structural studies have shown that the site that recognizes and binds to cell surface molecules during infection lies at the bottom of a surface cleft about 250 nm deep and 1.2 to 3 nm wide. Thus the binding site is well protected from the immune system while it carries out its functions, and the preparation of a vaccine directed against the binding site is not feasible. Possibly drugs that could fit in the cleft and interfere with virus attachment can be designed.

Viral invasion of the upper respiratory tract is the basic mechanism in the pathogenesis of a cold. The virus enters the body's cells by binding to the adhesion molecule ICAM-1 (**figure 38.14**). The clinical manifestations include the familiar stuffiness and/or partial obstruction, sneezing, scratchy throat, and a watery discharge from the nose. The discharge becomes thicker and assumes a yellowish appearance over several days. General malaise is commonly present. The disease usually runs its course in about a week. Diagnosis of the common cold is made from observations of clinical symptoms. There are no procedures for direct examination of clinical specimens or for serological diagnosis.

The source of the cold viruses may be infected individuals excreting viruses in nasal secretions, airborne transmission over short distances by way of moisture droplets, or transmission on contaminated hands or fomites. Epidemiological studies of rhinovirus colds have shown that the familiar explosive, noncontained sneeze (*see figure 37.6*) may not play an important role in virus spread. Rather, hand-to-hand contact between a rhinovirus “donor” and a susceptible “recipient” is more likely. The common cold occurs worldwide with two main seasonal peaks, spring and early autumn. Infection is most common early in life and generally decreases with an increase in age. Nothing is available for



(a)



(b)

Figure 38.14 Role of an Integrin (Inter cellular Adhesion Molecule-1) in Rhinovirus Uptake Into Cells of the Upper Respiratory Tract.

Rhinoviruses are a large family of viruses that are the single major cause of acute respiratory infections in humans. ICAM-1 (inter cellular adhesion molecule-1) has been identified as the cell surface receptor for the majority of these rhinoviruses. (a) In this computer-generated model, the virus-binding portion of ICAM-1 is shown in orange-red attaching to a rhinovirus (gray icosahedral protein capsid in the center). (b) This diagram indicates how the rhinovirus surface proteins (shown in blue-green) bind to the ICAM-1 molecule (red-yellow).

treating the common cold except additional rest, extra fluids, and the use of aspirin for alleviating local and systemic discomfort.

Cytomegalovirus Inclusion Disease

Cytomegalovirus inclusion disease is caused by the human cytomegalovirus (HCMV), a member of the family *Herpesviridae*. The virus contains a core with double-stranded DNA, an icosahedral capsid, and a phospholipid-rich envelope. Most people become infected with this virus at some time during their life; in the United States, as many as 80% of individuals older than 35 years have been exposed to this virus and carry a lifelong infection. Although most HCMV infections are asymptomatic, certain patient groups are at risk to develop serious illness and long-term effects from an HCMV infection. For example, this virus remains the leading cause of congenital virus infection in the United States, a significant cause of transfusion-acquired infections, and a frequent contributor to morbidity and mortality among organ transplant recipients and immunocompromised individuals (especially AIDS patients). Because the virus persists in the body, it is shed for several years in saliva, urine, semen, and cervical secretions.

The HCMV can infect any cell of the body, where it multiplies slowly and causes the host cell to swell in size—hence the prefix *cytomegalo*, which means “an enlarged cell.” Infected cells contain the unique **intranuclear inclusion bodies** and cytoplasmic inclusions (**figure 38.15**). In fatal cases, cell damage is seen in the gastrointestinal tract, lungs, liver, spleen, and kidneys. Normally cytomegalovirus inclusion disease symptoms resemble those of infectious mononucleosis.

Laboratory diagnosis is by viral isolation from urine, blood, lung, semen, or tissue. Serological tests (e.g., immunofluorescence, complement fixation, ELISA, and immunohistologic staining) and a rapid test kit (CMV-vue kit) also are available.

Epidemiologically the virus has a worldwide distribution, especially in developing countries where infection is universal by childhood. The prevalence of this disease increases with a lower-

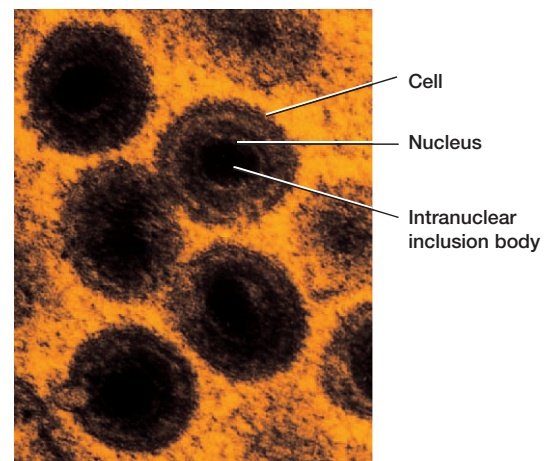
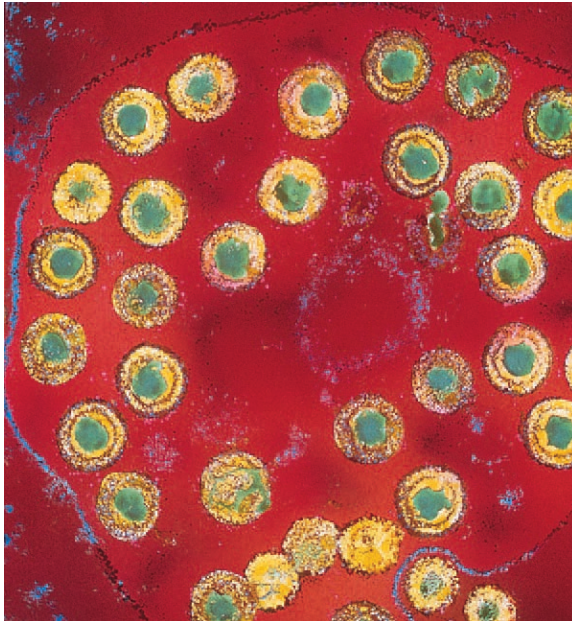


Figure 38.15 Cytomegalovirus Inclusion Disease. Electron micrograph of a single animal cell infected with the cytomegalovirus (×66,000). The intranuclear inclusion body has a typical “owl-eyed” appearance.

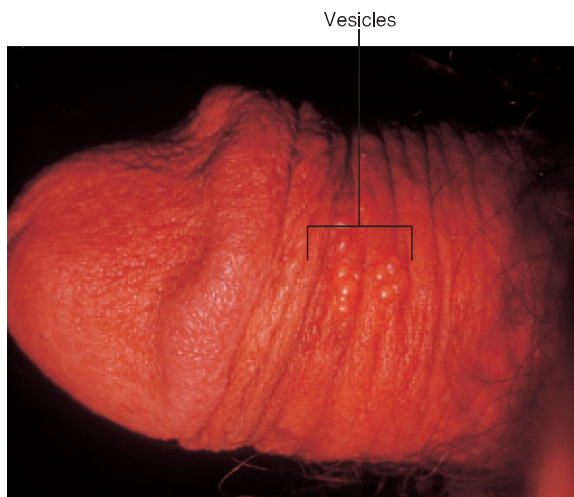
ing of socioeconomic status and hygienic practices. The only drugs available, ganciclovir (Cytovene-IV) and cidofovir (Vistide), are used only for high-risk patients. Infection can be prevented by avoiding close personal contact (including sexual) with an actively infected individual. Transmission by blood transfusion or organ transplantation can be avoided by using blood or organs from seronegative donors.

Genital Herpes

Genital herpes is caused by the herpes simplex virus type 2 (HSV-2). The HSV-2 virus (**figure 38.16a**) is classified in the alphaherpes subfamily of the family *Herpesviridae*. All members



(a)



(b)

Figure 38.16 Genital Herpes. (a) Herpes simplex virus type 2 (yellow and green) inside an infected cell (see also figure 16.17e). (b) Herpes vesicles on the penis. The vesicles contain fluid that is infectious.

of this subfamily have a very short replication cycle. The core DNA is linear and double stranded. The HSV-2 envelope contains at least eight glycoproteins. HSV-2 is most frequently transmitted by sexual contact (see table 39.4, p. 927 for a summary of all the sexually transmitted diseases covered in this textbook). Infection begins when the virus is introduced into a break in the skin or mucous membranes. The virus infects the epithelial cells of the external genitalia, the urethra, and the cervix. Rectal and pharyngeal herpes are also caused by sexual contact.

The HSV-2 genome must first enter an epithelial cell for the initiation of infection. The initial association is between the proteoglycans of the epithelial cell surface and viral glycoprotein C. This is followed by a specific interaction with one of several cellular receptors collectively termed HVEMs for “herpesvirus entry mediators.” The association also requires the specific interaction with glycoprotein D. Fusion with the epithelial cell plasma membrane follows. This requires the action of a number of other viral glycoproteins. The viral capsid along with some viral tegument proteins then migrates to nuclear envelope pores along the cellular microtubule transport machinery. This “docking” is thought to result in the viral DNA being injected through the nuclear envelope pores while the capsid remains in the cytoplasm. Some tegument proteins also enter the nucleus with the viral genome.

In the infected person, there is an active and latent phase. After an incubation period of about a week, the active phase begins. During the active phase, the virus multiplies explosively—between 50,000 and 200,000 new virions are produced from each infected cell. During this replication cycle, HSV-2 inhibits its host cell’s metabolism and degrades its DNA. As a result, the cell dies. Such an active infection may be symptom-free, or painful blisters in the infected tissue (figure 38.16b) may occur. The blisters are the result of cell lysis and the development of a local inflammatory response; they contain fluid and infectious viruses. A fever, headache, muscle aches and pains, a burning sensation, and genital soreness are frequently present during the active phase. Although blisters generally heal spontaneously in a few weeks, the viruses retreat to nerve cells in the sacral plexus, where they remain in a latent form. During the latent phase, the viral genome becomes incorporated into the nucleus of the host cell. During this phase, the host cell does not die. Because the viral genes are not expressed, the infected person is symptom-free. However, periodically the viruses multiply and migrate down nerve fibers to the skin or mucous membranes that the nerve supplies, where they produce new blisters. Activation may be due to sunlight, sexual activity, illness accompanied by fever, hormones, or stress. It should be noted that both the primary infection and reactivation can occur without any symptoms and apparently healthy people can transmit HSV-2 to their sexual partners or their newborns.

Besides being transmitted by sexual contact, herpes can be spread to an infant during vaginal delivery, leading to **congenital (neonatal) herpes**. Congenital herpes is one of the most life-threatening of all infections in newborns, affecting approximately 1,500 to 2,200 babies per year in the United States. It can result in neurological involvement as well as blindness. As a result any female who has had genital herpes should have a caesarean section instead of delivering vaginally. For unknown reasons, the HSV-2 virus is also associated with a higher-than-normal rate of cervical cancer and miscarriages.

Although there is no cure for genital herpes, oral use of the antiviral drugs acyclovir (Zovirax or Valtrex) and famciclovir (Famvir) has proven to be effective in ameliorating the recurring blister outbreaks. Topical acyclovir is also effective in reducing virus shedding, the time until the crusting of blisters occurs, and new lesion formation. Idoxuridine and trifluridine are used to treat herpes infections of the eye.

In the United States the incidence of genital herpes has increased so much during the past decade that it is now a very common sexually transmitted disease. It is estimated that over 25 million Americans (20% of adults) are infected with the herpes simplex virus type 2.

Human Herpesvirus 6 Infections

Human herpesvirus 6 (HHV-6) is the etiologic agent of **exanthem subitum** [Greek *exanthema*, rash] in infants. HHV-6 is a unique member of the family *Herpesviridae* that is distinct serologically and genetically from the other herpesviruses. The virus envelope encloses an icosahedral capsid and a core containing double-stranded DNA. The disease caused by HHV-6 was originally termed **roseola infantum** and then given the ordinal designation **sixth disease** to differentiate it from other exanthems and roseolas. Exanthem subitum is a short-lived disease characterized by a high fever of 3 to 4 days' duration, after which the temperature suddenly drops to normal and a macular rash appears on the trunk and then spreads to other areas of the body. HHV-6 infects over 95% of the United States infant population, and most children are seropositive for HHV-6 by 3 years of age. CD4⁺ T cells are the main site of viral replication, whereas monocytes are in an infected, latent state. The tropism of HHV-6 appears to be wide, including CD8⁺ T cells, natural killer cells, and probably epithelial cells. In adults HHV-6 is commonly found in peripheral-blood mononuclear cells and saliva, suggesting that the infection is lifelong. Since the salivary glands are the major site of latent infection, transmission is probably by way of saliva.

HHV-6 also produces latent and chronic infections and is occasionally reactivated in immunocompromised hosts leading to pneumonitis. Furthermore, HHV-6 has been implicated in several other diseases (lymphadenitis, multiple sclerosis, and infectious mononucleosis-like syndrome or chronic fatigue syndrome) in immunocompetent adults. Diagnosis is either by immunofluorescence or enzyme immunoassay. To date, there is no antiviral therapy or prevention.

Human Parvovirus B19 Infections

Since its discovery in 1975, **human parvovirus B19** (family *Parvoviridae*, genus *Parvovirus*) has emerged as a significant human pathogen. B19 virions are uniform, icosahedral, unenveloped particles approximately 23 nm in diameter. The genome of B19 is a single-stranded DNA. There is a spectrum of disease caused by parvovirus B19 infection, ranging from mild symptoms (fever, headache, chills, malaise) in normal persons, **erythema infectiosum** in children (**fifth disease**), and a joint disease syndrome in adults; to more serious diseases such as aplastic crisis in persons with sickle cell disease and autoimmune hemolytic anemia, and pure red cell aplasia due to persistent B19 virus infection in immunocompromised individuals. The B19 parvovirus can also infect the fetus, resulting in anemia and fetal hydrops (the accumulation of fluid in tissues). It is assumed that the natural mode of infection is by the respiratory route. A variety of techniques are available for the detection of the B19 virus. Antiviral antibodies appear to represent the principal means of

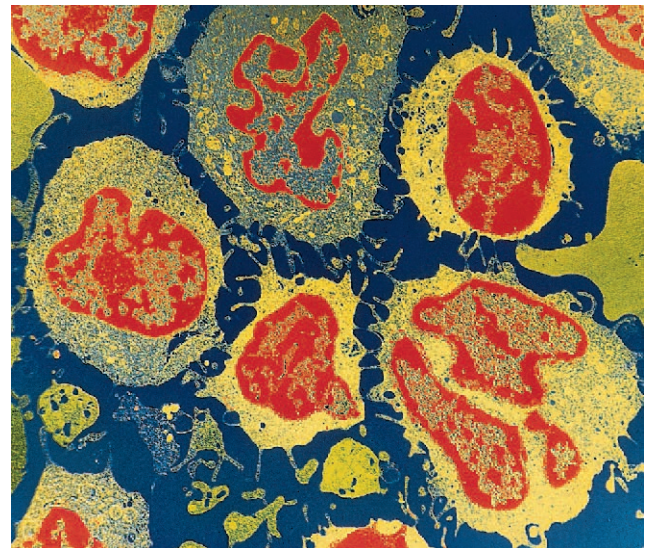


Figure 38.17 Hairy-Cell Leukemia. False-color transmission electron micrograph ($\times 3,100$) of abnormal B lymphocytes. Notice that the lymphocytes are covered with characteristic hairlike membrane-derived protrusions.

defense against B19 parvovirus infection and disease. The treatment of individuals suffering from acute and persistent B19 infections with commercial immunoglobins containing anti-B19 and human monoclonal antibodies to B19 have shown to be effective therapy. Most of the time infection is followed by lifelong immunity.

Leukemia

Certain **leukemias** in humans are caused by two retroviruses: human T-cell lymphotropic virus I (HTLV-I) and HTLV-II. HTLV-I and HTLV-II are members of the family *Retroviridae*. They have a nuclear core containing two positive-sense single-stranded RNA genomes. Once a cell is infected, the RNA genome is converted by reverse transcriptase to DNA and integrates into the host's genome. The viruses are transmitted by transfusions of contaminated blood, among drug addicts sharing needles, by sexual contact, across the placenta, from the mother's milk, or by mosquitoes. [Retroviruses and their replication](#) (p. 407); [Viruses and cancer](#) (pp. 411–12)

HTLV-1 causes **adult T-cell leukemia**. Once within the body the HTLV-I virus enters white blood cells and integrates into the cellular genome, where it activates the growth-promoting genes. The transformed cell proliferates extensively, and death generally results from the explosive proliferation of the leukemia cells or from opportunistic infections. To date, no effective treatment exists.

In 1982 the second human retrovirus (HTLV-II) was shown to be the agent responsible for hairy-cell leukemia. This virus shares the same disease-causing mechanism as HTLV-I. Hairy-cell leukemia gets its name from the many membrane-derived protrusions that give white blood cells the appearance of being "hairy" (**figure 38.17**). This leukemia is a chronic, progressive lymphoproliferative disease. The malignancy is believed to originate in a

stage of B-cell development. The bone marrow, spleen, and liver become infiltrated with malignant cells. This lowers the person's immunity. The primary cause of mortality is bacterial and other opportunistic infections. IFN- α n3 (Alferon N) has shown some promise for treatment in certain cases.

Mononucleosis (Infectious)

The Epstein-Barr virus (EBV), a member of the family *Herpesviridae*. As a member of the family *Herpesviridae*, EBV has the characteristic 120 nm enveloped morphology, with 162 capsomeres in icosahedral arrangement. Its double-stranded DNA exists both as a linear form in the mature virion and a circular episomal form in latently infected cells. EBV is the etiologic agent of **infectious mononucleosis (mono)**, a disease whose symptoms closely resemble those of cytomegalovirus-induced mononucleosis. Because the Epstein-Barr virus occurs in oropharyngeal secretions, it can be spread by mouth-to-mouth contact (hence the terminology infectious and kissing disease) or shared drinking bottles and glasses. Once a person has contracted mono, the virus enters lymphatic tissue, multiplies, and infects B cells. Infected B cells rapidly proliferate and take on an atypical appearance (Downey cells) that is useful in diagnosis. The disease is manifested by enlargement of the lymph nodes and spleen, sore throat, headache, nausea, general weakness and tiredness, and a mild fever that usually peaks in the early evening. The disease lasts for 1 to 6 weeks and is self-limited. Like other herpesviruses, EBV does become latent, probably in the B-cell population.

Treatment of mononucleosis is largely supportive and includes plenty of rest. Diagnosis of mononucleosis is made with a serological test for nonspecific (heterophile) antibodies. Several rapid tests are on the market.

The peak incidence of mononucleosis occurs in people 15 to 25 years of age. Collegiate populations, particularly those in the upper-socioeconomic class, have a high incidence of the disease. About 50% of college students have no immunity, and approximately 15% of these can be expected to contract mononucleosis. People in lower-socioeconomic classes tend to acquire immunity to the disease because of early childhood infection. The Epstein-Barr virus may well be the most common virus in humans as it infects 80 to 90% of all adults worldwide. EBV infections are associated with the cancers Burkitt's lymphoma in tropical Africa and nasopharyngeal carcinoma in Southeast Asia, East and North Africa, and in Eskimos.

Rabies

Rabies [Latin *rabere*, rage or madness] has been the object of human fascination, torment, and fear since the disease was first recognized. Prior to Pasteur's development of an antirabies vaccine, few words were more terrifying than the cry of "mad dog!" Improvements in prevention during the past 50 years have led to the almost complete elimination of indigenously acquired rabies in the United States. Worldwide, virtually all cases of human rabies are attributable to dog bites in developing countries where canine rabies is still endemic.

Rabies is caused by a number of different strains of highly neurotropic viruses. Most belong to a single serotype in the genus *Lyssavirus* [Greek *lyssa*, rage or rabies], family *Rhabdoviridae*. The bullet-shaped virion (**figure 38.18a**; see also figure 16.17c) contains a single-stranded, negative-sense RNA genome.

As previously noted, in the United States rabies is primarily a disease of animals. Most wild animals can become infected with rabies, but susceptibility varies according to species. Foxes, coyotes, and wolves are the most susceptible; intermediate are skunks, raccoons, insectivorous bats, and bobcats; and opossums are quite resistant.

The virus multiplies in the salivary glands of an infected host. It is transmitted to humans or other animals by the bite of an infected animal whose saliva contains the virus; by aerosols of the virus that can be spread in caves where bats roost; or by contamination of scratches, abrasions, open wounds, and mucous membranes with saliva from an infected animal.

After inoculation, a region of the virions' glycoprotein envelope spike attaches to the plasma membrane of nearby skeletal muscle cells, the virus enters the cells, and multiplication of the virus occurs. When the concentration of the muscle virus is sufficient, the virus enters the nervous system through unmyelinated sensory and motor terminals; the reported binding site is the nicotinic acetylcholine receptor. Since the virus is sequestered from the immune system, infection can no longer be halted by immunization. The virus spreads by retrograde axonal flow at 8 to 20 mm per day until it reaches the spinal cord, when the first specific symptoms of the disease—pain or paresthesia at the wound site—may occur. A rapidly progressive encephalitis develops as the virus quickly disseminates through the central nervous system. The virus then spreads throughout the body along the peripheral nerves, including those in the salivary glands, where it is shed in the saliva.

Within brain neurons the virus produces characteristic **Negri bodies**, masses of viruses or unassembled viral subunits that are visible in the light microscope. In the past the diagnosis of rabies consisted solely of examining nervous tissue for the presence of these bodies. Today diagnosis is based on direct immunofluorescent-antibody (dIFA) of brain tissue, virus isolation, detection of Negri bodies, and a rapid rabies enzyme-mediated immunodiagnosis test.

Symptoms of rabies usually begin 2 to 16 weeks after viral exposure and include anxiety, irritability, depression, fatigue, loss of appetite, fever, and a sensitivity to light and sound. The disease quickly progresses to a stage of paralysis. In about 50% of all cases, intense and painful spasms of the throat and chest muscles occur when the victim swallows liquids. The mere sight, thought, or smell of water can set off spasms. Consequently rabies has been called hydrophobia (fear of water). Death results from destruction of the regions of the brain that regulate breathing.

Safe and effective vaccines (human diploid-cell rabies vaccine HDCV [Imovax Rabies] or rabies vaccine adsorbed [RVA]) against rabies are available; however, to be effective they must be given soon after the person has been infected. Veterinarians and laboratory personnel, who have a high risk of exposure to rabies, usually are immunized every 2 years and tested for the presence of suitable antibody titer. About 30,000 people annually receive this treatment. In the United States fewer than 10 cases of rabies

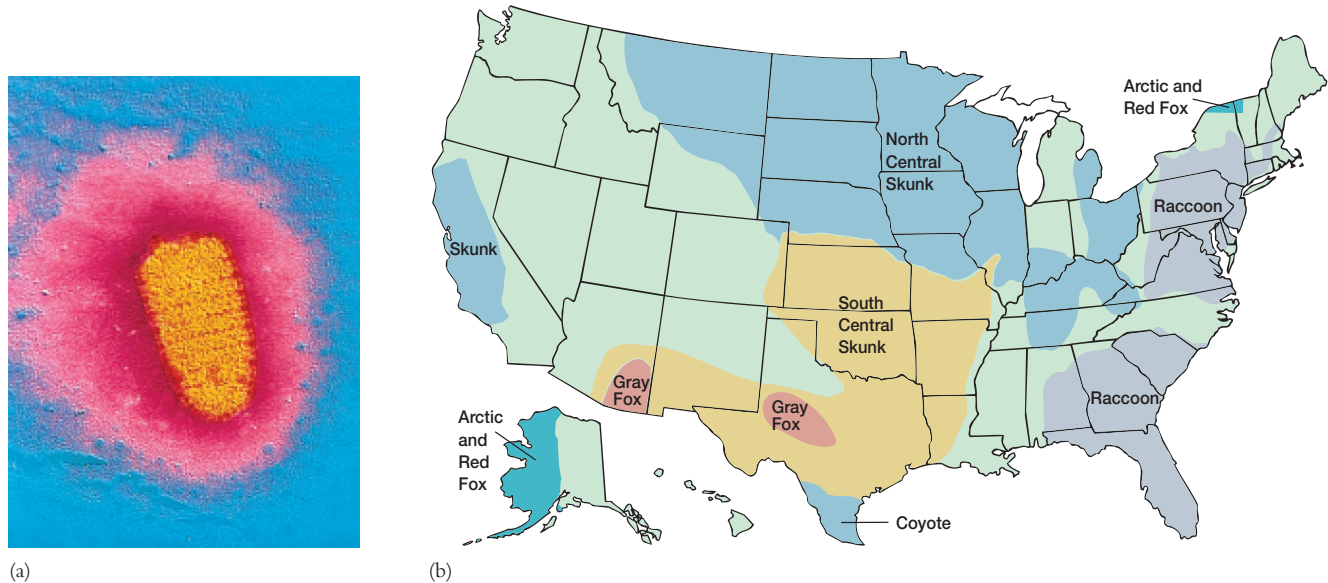


Figure 38.18 Rabies. (a) Electron micrograph of the rabies virus (yellow) ($\times 36,700$). Note the bullet shape. The external surface of the virus contains spikelike glycoprotein projections that bind specifically to cellular receptors. (b) In the United States rabies is found in terrestrial animals in 10 distinct geographic areas. In each area a particular species is the reservoir and one of five antigenic variants of the virus predominates as illustrated by the five different colors. Although not shown, another eight viral variants are found in insectivorous bats and cause sporadic cases of rabies in terrestrial animals throughout the country. Absence of a strain does not imply absence of rabies.

occur yearly in humans; about 8,000 cases of animal rabies are reported each year from various sources (figure 38.18b). Prevention and control involves preexposure vaccination of dogs and cats, postexposure vaccination of humans, and preexposure vaccination of humans at special risk (persons spending a month or more in countries where rabies is common in dogs).

Viral Hepatitides

Any infection that results in inflammation of the liver is called **hepatitis** [pl., hepatitises] [Greek *hepaticus*, liver]. Currently nine viruses are recognized as causing hepatitis. Two are herpesviruses (cytomegalovirus [CMV] and Epstein-Barr virus [EBV]) and seven are hepatotropic viruses.

EBV and CMV cause mild, self-resolving forms of hepatitis with no permanent hepatic damage. Both viruses cause the typical infectious mononucleosis syndrome of fatigue, nausea, and malaise.

Of the eight human hepatotropic viruses, only five are well characterized; hepatitis F, G, and TTV (transfusion-transmitted virus) are newly discovered viruses (table 38.4). Hepatitis A (sometimes called infectious hepatitis), and hepatitis E (formerly enteric-transmitted NANB hepatitis), are transmitted by fecal-oral contamination and discussed in the section on food- and water-borne diseases (section 38.4). The other major types include hepatitis B (sometimes called serum hepatitis), hepatitis C (formerly non-A, non-B hepatitis), and hepatitis D (formerly delta hepatitis).

Hepatitis B (serum hepatitis) is caused by the hepatitis B virus (HBV), a double-stranded circular DNA virus of complex structure. HBV is classified as an *Orthohepadnavirus* within the family *Hepadnaviridae*. Serum from individuals infected with hepatitis B contains three distinct antigenic particles: a spherical 22 nm particle, a 42 nm spherical particle (containing DNA and DNA polymerase) called the **Dane particle**, and tubular or filamentous particles that vary in length (figure 38.19). The small spherical and tubular particles are the unassembled components of the Dane particle—the infective form of the virus. The unassembled particles contain **hepatitis B surface antigen** (HBsAg) whose presence in the blood is (1) an indicator of hepatitis B infection, (2) the basis for the large-scale screening of blood for the hepatitis B virus, and (3) the basis for the first vaccine for human use developed by recombinant DNA technology.

The hepatitis B virus is normally transmitted through blood transfusions, contaminated equipment, drug users' unsterile needles, or any body secretion (saliva, sweat, semen [see table 39.4, p. 927], breast milk, urine, feces). The virus also can pass from the blood of an infected mother through the placenta to infect the fetus. Each year an estimated 300,000 people in the United States are infected with HBV. About 5,000 persons die yearly from hepatitis-related cirrhosis and about 1,000 die from HBV-related liver cancer. (HBV is second only to tobacco as a known cause of human cancer.) Worldwide, HBV infects over 200 million people.

Table 38.4 Characteristics of Hepatitides Caused by Hepatotropic Viruses^a

Disease (Virus)	Genome	Classification	Transmission	Outcome	Prevention
Hepatitis A (Hepatitis A)	RNA	<i>Picornaviridae</i> , <i>Hepatovirus</i>	Fecal-oral	Subclinical, acute infection	Killed HAV (Havrix vaccine)
Hepatitis B (Hepatitis B)	DNA	<i>Hepadnaviridae</i> , <i>Orthohepadnavirus</i>	Blood, needles, body secretions, placenta, sexually	Subclinical, acute chronic infection; cirrhosis; primary hepatocarcinoma	Recombinant HBV vaccines
Hepatitis C (Hepatitis C)	RNA	<i>Flaviviridae</i> , <i>Pestivirus</i> , or <i>Flavivirus</i> (?)	Blood, sexually	Subclinical, acute chronic infection; primary hepatocarcinoma	Routine screening of blood
Hepatitis D (Hepatitis D)	RNA	Unclassified	Blood, sexually	Superinfection or coinfection with HBV	HBV vaccine
Hepatitis E (Hepatitis E)	RNA	<i>Caliciviridae</i> (?)	Fecal-oral	Subclinical, acute infection (but high mortality in pregnant women)	Improve sanitary conditions

^aHepatitis F, G, and TTV have been discovered but not well characterized. Thus they are not included in this table.

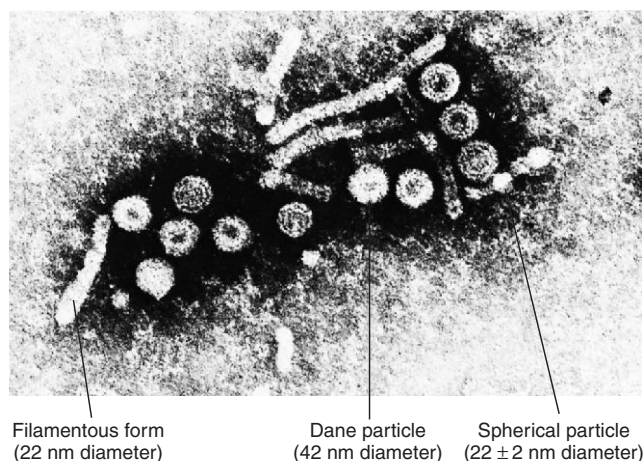


Figure 38.19 Hepatitis B Virus in Serum. Electron micrograph ($\times 210,000$) showing the three distinct types of hepatitis B antigenic particles. The spherical particles and filamentous forms are small spheres or long filaments without an internal structure, and only two of the three characteristic viral envelope proteins appear on their surface. Dane particles are the complete, infectious virion.

The clinical signs of hepatitis B vary widely. Most cases are asymptomatic. However, sometimes fever, loss of appetite, abdominal discomfort, nausea, fatigue, and other symptoms gradually appear following an incubation period of 1 to 3 months. The virus infects liver hepatic cells and causes liver tissue degeneration and the release of liver-associated enzymes (transaminases) into the bloodstream. This is followed by jaundice, the accumulation of bilirubin (a breakdown product of hemoglobin) in the skin and other tissues with a resulting yellow appearance. Chronic hepatitis B infection also causes the development of primary liver cancer, hepatocellular carcinoma

(hepatitis B is second in importance only to tobacco as a known human carcinogen).

General measures for prevention and control involve (1) excluding contact with HBV-infected blood and secretions, and minimizing needle-sticks by scrupulous technique; (2) passive prophylaxis with intramuscular injection of hepatitis B immune globulin within 7 days of exposure; and (3) active prophylaxis with two recombinant vaccines: Engerix-B, and Recombivax HB. These vaccines are widely used by health professionals who are at increased risk of contacting the hepatitis B virus. Other individuals who should be vaccinated include:

1. Contacts of HBV carriers (e.g., household members, sex partners, institutional inmates)
2. International travelers (*see table 37.5*)
3. Sexually active homosexual males
4. Hemodialysis patients
5. Recipients of blood and related products that are possibly HBV contaminated

Another form of hepatitis is called **hepatitis C**. The virion has an 80 nm diameter, has a lipid coat, contains a single strand of linear ssRNA, and has been designated the hepatitis C virus (HCV) within the family *Flaviviridae*. HCV has recently been classified into multiple genotypes. This virus is transmitted by intimate contact with virus-contaminated blood, by the fecal-oral route, by in utero transmission from mother to fetus, sexually, or through organ transplantation. Diagnosis is made by a first-generation enzyme-linked immunosorbent assay (ELISA), which detects serum antibody to a recombinant antigen of HCV. Hepatitis C is found worldwide. In recent years hepatitis C has accounted for more than 90% of hepatitis cases developing after a blood transfusion. Worldwide, hepatitis C has reached epidemic proportions, with more than 1 million new cases reported annually. In the United States alone, nearly 4 million persons are infected and 30,000 new cases occur annually. Currently, HCV is responsible for 8,000 to 10,000 deaths

annually in the United States, and without effective intervention, that number is predicted to triple in the next 10 years. Furthermore, HCV is the leading reason for liver transplantation in the United States. Treatment is with recombinant IFN- α 2b or IFN- α 2a (Intron A, Roferon-A) three times weekly for 6 months.

In 1977 a cytopathic hepatitis agent termed the **Delta agent** was discovered. Later it was called the hepatitis D virus (HDV) and the disease **hepatitis D** designated. HDV is a unique agent in that it is dependent on the hepatitis B virus to provide the envelope protein (HBsAg) for its RNA genome. Thus HDV only replicates in liver cells in which the HBV also is actively replicating. Furthermore, the RNA of the HDV is smaller than the RNA of the smallest picornaviruses and its circular conformation differs from the linear structure typical of animal RNA viruses. (The nearest relatives of the HDV are in the plant kingdom—the viroids, virusoids, and satellite RNAs [see chapter 18].) HDV is spread only to persons who are already infected with HBV (superinfection) or to individuals who get both viruses at once (coinfection). The primary laboratory tools for the diagnosis of an HDV infection are serological tests for anti-delta antibodies. Treatment of patients with chronic HDV remains difficult. Early studies indicate that some positive results can be obtained with alpha interferon for 3 months to 1 year. Worldwide, there are approximately 300 million HBV carriers, and available data indicate that no fewer than 5% of these are infected with HDV. Thus because of the propensity of HDV to cause fulminant as well as chronic liver disease, continued incursion of HDV into areas of the world where persistent hepatitis B infection is endemic will have serious implications. Prevention and control involves the widespread use of the hepatitis B vaccine.

Recently the presence of two other forms of hepatitis have been identified: **hepatitis F** (causing fulminant, post-transfusion hepatitis) and a syncytial giant-cell hepatitis (**hepatitis G**) with viruslike particles resembling the measles virus (*Paramyxovirus*). Further virologic, epidemiological, and molecular efforts to clone and characterize these new agents and their diseases are currently being undertaken.

1. Describe the AIDS virus and how it cripples the immune system. How is the virus transmitted? What types of pathological changes can result?
2. Why do people periodically get cold sores? Describe the causative agent.
3. Why do people get the common cold so frequently? How are cold viruses spread?
4. Give two major ways in which herpes simplex virus type 2 is spread. Why do herpes infections become active periodically?
5. What two types of leukemias are caused by viruses?
6. Describe the causative agent and some symptoms of mononucleosis and exanthem subitum.
7. How does the rabies virus cause death in humans?
8. What are the different causative viruses of hepatitis and how do they differ from one another? How can one avoid hepatitis? Do you know anyone who is a good candidate for infection with these viruses?

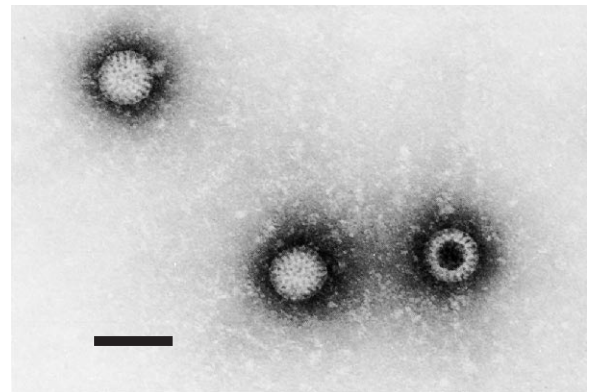


Figure 38.20 Viral Gastroenteritis. Electron micrograph of rotaviruses (reoviruses) in a human gastroenteritis stool filtrate. Note the spokelike appearance of the icosahedral capsids that surround double-stranded RNA within each virion. Bar = 100 nm.

38.4 Food-Borne and Waterborne Diseases

Food and water have been recognized as potential carriers of disease since the beginning of recorded history. Collectively more infectious diseases occur by these two routes than any other. A few human viral diseases that are food- and waterborne are now discussed. [Water-based diseases \(section 29.5\); Diseases and food \(section 41.4\)](#)

Gastroenteritis (Viral)

Acute viral gastroenteritis (inflammation of the stomach or intestines) is caused by four major categories of viruses: rotaviruses (**figure 38.20**), Norwalk and Norwalk-like viruses, other caliciviruses, and astroviruses. The medical importance of these viruses is summarized in **table 38.5**.

The viruses responsible for gastroenteritis are probably transmitted by the fecal-oral route. Infection is most common during the cooler months in contrast to bacteria-caused diarrheal diseases, which usually occur in the warmer months of the year. Diarrheal diseases are the leading cause of childhood deaths (5 to 10 million deaths per year) in developing countries where malnutrition is common. Current estimates are that viral gastroenteritis produces 30 to 40% of the cases of infectious diarrhea in the United States, far outnumbering documented cases of bacterial and parasitic diarrhea (the cause of approximately 40% of presumed cases of diarrhea remains unknown). In the United States rotaviruses account for about 3.5 million cases of illness, resulting in 35% of the hospitalizations for gastroenteritis, and 75 to 150 deaths each year.

Viral gastroenteritis is seen most frequently in infants 1 to 11 months of age, where the virus attacks the upper intestinal epithelial cells of the villus, causing malabsorption, impairment of